

Reliability of Physics-Transferred Neural Surrogates in Gradient-Based Inverse Metasurface Design: A Pre-Specified Reference-Solver Audit

Sang-Bae Choi¹, Je-Min Choi², Joonhyub Kim², Chang-Mo Kang²

¹*Independent Researcher, Busan, 46264, Republic of Korea*

²*Department of Nanomechatronics Engineering, Pusan National University, Busan, 46241, Republic of Korea*

Corresponding authors: Sang-Bae Choi (sbchoi129@gmail.com) and Chang-Mo Kang

Abstract

Physics-based transfer learning produces data-efficient neural-network surrogates for nanophotonic absorbers, and a companion study introduced a pilot-set transferability diagnostic, the shape correlation r between a cheap and an expensive solver read together with their amplitude error, that anticipates the forward gain. We ask whether it also certifies such a surrogate as the forward block of a gradient-based inverse-design loop. We froze a reliability protocol, its thresholds and three hypotheses before the main runs, and ran it on the published pipeline of that study: three metal–insulator–metal absorber structures at three training seeds each, with a rigorous coupled-wave reference solver called once per committed geometry to count pretenders, the designs the surrogate certifies but the solver rejects. All 179 valid committed designs pass the surrogate's 5 % check and the solver rejects 16 (8.9 %; 95 % CI [4.0, 14.5] target-clustered, [5.6, 14.0] nominal; one further solve failed), so the pass verdict screened none of them out. At 2.5 % tolerance 53 of the 171 designs still certified are pretenders (31 %), at 7.5 % 6 of 179. None of three tested factors is individually necessary. A surrogate trained from scratch still produces pretenders (6 $\rightarrow$ 4 at the pre-specified seed, 16 $\rightarrow$ 17 over three seeds), confining the optimizer to the generator-feasible region reduces but does not remove them (6 $\rightarrow$ 3; 16 $\rightarrow$ 5), and budget-matched random search produces them too. What every control retains is committing to the argmin of a surrogate that is imperfect where it is most optimistic. A fourth structure audited out of sample gives 4 of 50, inside the band recorded for it. The identical protocol on the release this pipeline superseded gives 95 of 179 (53.1 %). Such reliability claims therefore attach to a release rather than to a method.

Keywords: neural surrogate; inverse design; transfer learning; metasurface absorber; offline model-based optimization; reliability audit; pre-specified protocol

1 Introduction

1.1 Problem statement

Learned surrogates are now routinely placed inside optimization loops across the computational sciences: a network is trained to imitate an expensive simulator, and a designer then optimizes against the network because the simulator is too slow to query in the loop. The loop consumes the surrogate's gradients, not its average accuracy, and it actively seeks the points where the surrogate is most optimistic. Yet the decision to trust a surrogate in that role is commonly made on a forward metric alone — a test error, or a transferability score computed before training — and the reliability of that decision is rarely audited, still more rarely under a protocol fixed in advance.

Our testbed is nanophotonic absorber design, where surrogates are standard equipment and several reviews survey the field [3, 4, 5, 6, 7]. Metal–insulator–metal absorbers [2] are the structure family, and the companion study [1] supplies the surrogates: it pre-trains on a cheap transfer-matrix solver, fine-tunes on a rigorous coupled-wave one, and introduces a pilot-set diagnostic — the shape correlation r between the two solvers, read jointly with the operating-band amplitude error — that anticipates which structures gain from the transfer. Transfer between physical scenarios [8], design tasks [9] and physical priors [10, 11] has precedent in this field, as does negative transfer [12], which is why such a diagnostic exists at all.

The question this paper asks is the general one: does a forward transferability diagnostic certify a surrogate for *inverse* use? We treat it as falsifiable, fix the protocol, its thresholds and three ordered

hypotheses (H1a–c, §1.4) before the main runs, and run three structures spanning the diagnostic's range through one codebase at three training seeds each.

1.2 Forward → inverse risk asymmetry

Forward surrogate prediction is a local interpolation problem: a small surrogate error contributes additively to the average MAE. Inverse optimization is categorically different. The optimizer $x^* = \arg \min_x \mathcal{L}(f_{NN}(x), y_{\text{target}})$ *actively searches* for geometries that minimize the surrogate-reported loss, so wherever the surrogate's error landscape contains a basin shallower than the true loss basin, the optimizer preferentially steps into it. A 3 % forward MAE thus becomes, in the inverse direction, an *exploitation target* rather than an additive noise floor. This is the same pathology that the offline model-based-optimization literature calls oracle gaming or surrogate exploitation [13, 14, 15, 16]: optimizing against a fixed learned objective drives the solution into regions where the objective is over-optimistic. Forward metrics (MAE, R^2 , spectrum match) are therefore necessary but not sufficient indicators of inverse usability, and an optimizer-aware, oracle-in-the-loop reliability framework is required.

1.3 Related work

Managing an approximate model inside an optimization loop is an old problem. Trust-region model management [17] and the surrogate-management framework [18] let a cheap model drive the search while evaluations of the expensive function decide whether each step is accepted and keep the iteration convergent, and the surrogate-based and surrogate-assisted optimization literatures have developed that model-management bargain since [19, 20]. Multi-fidelity modelling treats the same two-solver structure explicitly — an autoregressive or composite model relates a cheap low-fidelity source to a scarce high-fidelity one [21, 22, 23], and sequential multi-fidelity optimisation goes further by choosing, per query, which fidelity to evaluate [60] — and we read the TMM→RCWA transfer audited here as a multi-fidelity scheme in which the cheap solver enters through pre-trained weights rather than through an explicit discrepancy term; that reading is ours, not a result of those works. What these sequential model-management methods have in common, and what a forward transferability score lacks, is a step at which the expensive model is consulted about the cheap model's answer. The protocol below keeps only that step, in its cheapest form — one high-fidelity call at the committed point — and gives up the iterative guarantees that [17, 18] obtain by consulting the expensive model throughout.

The failure mode we measure is the one offline model-based optimization has named. Optimizing against a fixed learned objective drives the solution into regions where that objective is over-optimistic, and the remedies proposed respond to it in different ways: conditioning by adaptive sampling [24] and weighted retraining [25] adapt the proposal or generative distribution toward designs the data support, model inversion networks [13] learn the map from objective value to design directly, and autofocused oracles [16] and robust model adaptation [26] re-weight or adapt the predictive model around the designs the search proposes. Design-Bench provides a standardised benchmark suite for the setting [15], and a recent review surveys the field [27]. The same structure appears wherever a proxy is optimized — regressional and extremal Goodhart effects [28], reward hacking [29] — and best-of- n selection against a learned reward model is the sharpest analogy to our commitment rules: Gao et al. [30] show that selecting the best of n samples by a proxy can raise the proxy score while the gold score falls. Choosing the argmin of an imperfect surrogate over eight restarts, or over 6400 random draws, is the same operation, which is why budget-matched random search produces pretenders as gradient descent does (Section 3.6). In silico sequence-design benchmarking has independently reported that learned oracles used to score generated candidates disagree with one another by architecture and even by training seed, and generalise poorly to the out-of-distribution sequences that design methods propose [31].

The companion study itself recorded a preliminary inverse-design probe on its earlier visible-band Structure-A surrogate ([1], Supplementary S13): gradient optimization exploited the surrogate and the committed designs failed under two RCWA solvers. That qualitative observation is the antecedent of this audit, which turns it into a pre-specified, multi-structure, multi-seed accounting on the published pipeline.

Two responses to an unreliable surrogate must be distinguished. (i) *Surrogate-improving* calls spend the simulation budget on the model: active learning selects the expensive simulations that most improve the surrogate [32], and physics-driven training folds the solver into the training objective of a design-generating network [33]. (ii) *Certification-only* calls spend the budget on checking the committed answer and change the model not at all; this is the protocol of this paper, and it differs from the iterative model management of [17, 18] in consulting the expensive model once, at the end. In our judgement the first is the better long-run investment and the second is what a practitioner who has already trained a surrogate can do today; its cost is one full-spectrum solve per design by construction of the protocol (103–4595 s here, Section 3.5). This paper measures how much that single call buys.

Finally, the reliability of the claim itself. Pre-registration separates confirmatory from exploratory analysis [34] and has recently been argued for predictive modelling specifically [35]; reproducibility programmes ask for the code and artifacts that let a reader re-derive a number [36]; and overoptimistic claims in machine-learning-based science have been traced to leakage across many fields [37] and, for machine learning applied to fluid-related partial differential equations, to weak baselines and reporting biases [38]. We follow that line by freezing the protocol, its thresholds and its hypotheses before the main runs, labelling every post-hoc analysis as such, and releasing the artifact behind every number.

1.4 Hypotheses and pre-specification

The protocol (reproduced in Supplementary S4 as preserved, with the two items added after the freeze marked there) was frozen on 2026-06-10, before the seed-42 runs. It states the research question, the metric hierarchy, the taxonomy, every threshold, the target count, the seeds, and the statistical analysis plan. Three pre-specified orderings test whether downstream reliability degrades as r falls, across the three structures A (median $r = +0.72$), C (+0.34), B (−0.07) — the *preliminary* pilot values available when the protocol was frozen (2026-06-10), superseded by the published Table 5 of [1], which gives B +0.96, A +0.83, C +0.65 (Section 2.1, W1). The protocol keyed the orderings to the shape correlation alone; the printed study reads r jointly with the operating-band MAE, so we report that ordering as well (Section 3.3). All three conventions are analysed:

- **H1a** — oracle-confirmed success rate is non-increasing as r falls ($A \geq C \geq B$).
- **H1b** — Δ -flag rate is non-decreasing as r falls ($A \leq C \leq B$).
- **H1c** — $\rho(\text{RCWA MAE}, \text{Surr MAE})$, the informativeness of the surrogate's self-report about true quality, is non-increasing as r falls.

Any outcome is reportable. Confirmation makes the audit a direct sequel to the diagnostic of [1]; non-support is itself a finding — the forward diagnostic would then be shown not to supply, by itself, a demonstrated inverse guarantee, leaving the protocol as the contribution.

1.5 Contributions

1. **A pre-specified reliability protocol** (Sections 2.3–2.5): a formal pretender definition, a reference-solver gap audit, a fake-optimum taxonomy with every test measured, calibrated random nulls and a full analysis plan, frozen beforehand and reusable on any surrogate-in-the-loop design pipeline.
2. **A non-certification result for the threshold check** (Sections 3.2–3.4): every valid committed geometry passes the surrogate's own 5 % check, and the solver rejects 16 of 179 — so on these designs the pass/fail verdict screened out no solver failure, while the *continuous* self-report can still discriminate on some structures (Section 3.7). How many designs the solver rejects depends on the tolerance demanded: 8.9 % at $\tau = 5$ %, 31 % at $\tau = 2.5$ %.
3. **Three controls that rule out the usual suspects as necessary causes** (Sections 3.6, 3.8 and Table 9): pretenders persist without transferred weights, without the gradient loop and without geometric infeasibility. The step every control keeps is committing to the argmin of a surrogate that is imperfect where it is most optimistic.

4. **A release-dependence result** (Section 3.10): the identical protocol on the release this pipeline superseded gives 53.1 % against 8.9 %, so a reliability claim of this kind characterizes a specific release rather than a structure or a method — which is an argument for auditing the pipeline one actually ships.
5. **An out-of-sample structure** (Section 3.12): a fourth absorber generated after the protocol was frozen, audited on its full 50-design held-out split against a rate band recorded before its runs, gives 4 of 50 (8.0 %) — inside the band, with every pretender again a geometry the optimizer pushed outside the generator’s feasible region.

Where the diagnostic *does* carry signal, and where the cheap detectors do, is reported with its limits rather than as a headline: pretender incidence follows the pre-specified r ordering across these structures (Section 3.3) and two of the oracle-free detectors reach an AUROC near 0.8 or above on two of them (Section 3.7), both on event counts too small to establish a rule.

Two qualifications travel with every number above. The verdicts are conditional on the pinned reference solver, whose own raster and truncation limits are measured in Section 4.7 (W8); and the Structure-D prediction record, written before its runs, was amended in its count conversion and falsification bound (the 5–15 % band itself unchanged) while partial oracle results existed on disk, uninspected (Section 3.12).

The operational form is the checklist of Box 1 (Section 4.4).

2 Methods

2.1 Surrogates and the three structures

All three surrogates share the M_TL+phys architecture of [1] (ResNet-256-4 backbone + Sigmoid head, mapping a normalized wavelength concatenated with normalized geometry parameters and structure-specific physics features to absorbance). The three structures span the r range and differ in dimensionality and output channels:

Table 1: The three absorber structures (design dimensionality, physics-feature count, wavelength grid, filtered dataset and surrogate head; columns ordered by the published r ; the r values, preliminary r values and filtered yields are published measurements of [1] — Table 5, Supplementary S19 and Methods — restated here as the inputs to this audit). The main text audits the published pipeline of [1]; the as-submitted release, whose datasets, bands and feature sets differ, is the second arm of Section 3.10 and is described in Supplementary Section S8.

	Structure B ($r = +0.96$; prelim. -0.07)	Structure A ($r = +0.83$; prelim. $+0.72$)	Structure C ($r = +0.65$; prelim. $+0.34$)
Physics	ring-disk Pano	asym. dual-dielectric dual-cavity	anisotropic rect. (TE/TM)
Design params d	8	10	7
Physics features	13	17	18
Wavelength grid	400–1800 nm $\times$ 100	400–1800 nm $\times$ 100	400–1800 nm $\times$ 100
Dataset (reliable/500)	struct_B_500_redesign (498)	struct_A_500_redesign (499)	struct_C_500_redesign (487)
Output channels	A, R	A, R	A_TE,R_TE,A_TM,R_TM
Surrogate head	single	single	dual

Each surrogate is fine-tuned from a TMM-pretrained M_TL+phys checkpoint, loaded **strict**. The published pipeline of [1] does not ship its checkpoints, so they were regenerated here with its own pretraining recipe (pbt1_{A,B}_redesign.py, pbt1_C_v2_redesign.py, steps 1–2 at commit cb486b5: 5000 TMM spectra drawn with rng(99), 500 epochs, lr 10^{-3} , batch 2048, seed 42) and archived with their hashes (Table 2). Faithful loading requires: (i) geometry input ordered [wl_norm, params_norm] (wavelength first); (ii) params_norm using the **training bounds of the checkpoint**, which in this pipeline coincide with the dataset sampling box (for C, $W_x, W_y \in [50, 720]$ nm; the as-submitted release trained C on a narrower [50, 400] box than it sampled, see Section 4.7); (iii) physics features normalized with the **TMM-set statistics**, recomputed deterministically; and (iv) the upstream recipe (MSE over all output channels, AdamW [39] lr 3×10^{-4} , weight decay 10^{-4} , cosine annealing, 1000 epochs, batch 512, best-val checkpoint, seed 42). Data hygiene and the train/val/test split mirror [1] exactly: the pipeline’s

`reliable` mask selects 499 / 498 / 487 of the 500 generated samples for A / B / C, A and R are clipped to [0,1], and a `rng(42)` permutation assigns `test` = last 50, `val` = previous 50 and the fine-tune train set = first 350 of the permuted remainder (A 350/399, B 350/398, C 350/387). The same pipeline is run at training-subset/training seeds 123 and 777 for the multi-seed extension.

Data, checkpoint and solver provenance. The audit is pinned to one version of [1]. The audited inputs are, by SHA-256 prefix (full digests in `INPUTS_MANIFEST.txt` in the repository),

Table 2: Provenance of the audited inputs (SHA-256 prefixes of the filtered dataset and of the TMM-pretrained checkpoint that every run starts from).

structure	dataset (SHA-256)	TMM-pretrained checkpoint (SHA-256)
B	a75c4de5...	892aabda...
A	9dea0fa6...	66aa7db1...
C	57bb2b99...	84fe0a59...

The three dataset digests equal those listed for `struct_{A,B,C}_500_redesign.npz` in the companion's own data manifest at the same commit, so the audited data are the files behind its printed tables, not a regeneration. All simulation code is vendored at upstream commit `cb486b5`, the commit behind the printed tables of [1]. Its solver settings are a 64×64 real-space grid with a **per-wavelength adaptive Fourier order** — the truncation is set to $N = 9, 13$ or 17 by a fixed rule of the smallest metal feature, the patch aspect ratio and the wavelength for Structures A and C, of the smallest feature and air gap alone for Structure D, or of the period-to-wavelength ratio for Structure B ([1], Supplementary S16; it is a heuristic, not a per-solve convergence test). The raster's Fourier couplings are free of index wrap-around only up to $N = 15$ (Section 4.7, W8), and Section 3.5 reports the orders it actually chose — in complex64 arithmetic, with Johnson & Christy tabulated constants for Cr, a Malitson Sellmeier model for SiO_2 and the measured Siefke (2016) constants for TiO_2 . The as-submitted release (commit `920b1bd`) ran a fixed Fourier order 5 in complex128 with a Devore Sellmeier model for TiO_2 , on datasets with different bands and hygiene; numbers from the two pipelines are therefore not interchangeable, every comparison in Section 3.1 is made against the published tables, and the two arms are compared as two independent runs of the identical protocol in Section 3.10 — the design-by-design comparison of the solver alone is Section 3.11.

2.2 Inverse-design pipeline

For each target absorptance spectrum $A^*(\lambda)$ over 100 wavelength channels (C: TE and TM jointly) we solve (Figure 1)

$$u^* = \arg \min_{u \in [0,1]^d} \frac{1}{100} \sum_{\lambda} (f_{NN}(g(u), p(g(u), \lambda)) - A^*(\lambda))^2,$$

where the optimization variable u is the **normalized** geometry, mapped to physical units $g(u)$ on the dataset sampling box and **clamped to [0, 1] after every step**. We use Adam [41] on u , lr 0.05×500 iters then 0.02×300 iters, with **$R = 8$ multi-start restarts** per target (one mid-box draw plus seven uniform draws); all eight runs proceed as one batched tensor, the committed geometry is the restart with the lowest final surrogate loss, and all eight endpoints are archived for the Tier-1C dispersion diagnostic. Physics features are recomputed differentially from the current geometry at every iteration with the §2.1 TMM normalization.

Targets. $N = 20$ per structure, drawn `rng(42).choice(test_split, 20, replace=False)` — random, from held-out data, by the **identical** mechanism for all three structures. (Restart seeds use `rng(1000 + original-dataset index of the target)`, a fixed pre-data quantity; recorded for exact reproducibility.)

2.3 Reliability metric hierarchy

Tier 1A — success rate and pretenders. At threshold $\tau = 5\%$: a design is *surrogate-claimed* if Surr MAE $\leq \tau$ and *oracle-confirmed* if RCWA MAE $\leq \tau$. A **pretender** is a design with Surr MAE $\leq \tau$ and RCWA MAE $> \tau$ (definition fixed before any computation). Wilson 95 % CIs [42] are reported on all counts, including zeros.

Tier 1B — oracle revalidation gap. $\Delta_i = \text{MAE}(A_{\text{RCWA}}(x_i^*), A_i^*) - \text{MAE}(f_{\text{NN}}(x_i^*), A_i^*)$, flagged when $\Delta_i > k \cdot \text{MAE}_{\text{NN}, \text{test}}$ with $k = 2$. We report max Δ and, only with the small-denominator caveat, the amplification RCWA/Surr.

Tier 1C — convergence. Multi-start dispersion (std of the 8 final losses; max pairwise endpoint distance in u-space) plus the loss-tail RSD of the committed run.

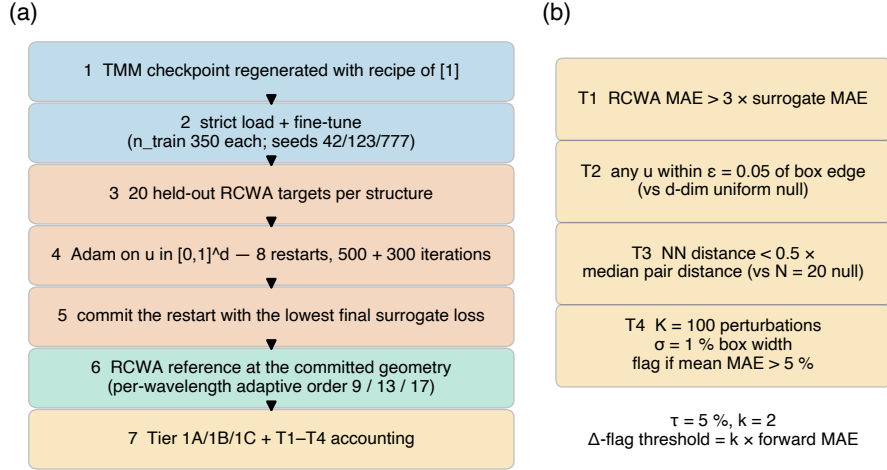


Figure 1: Protocol overview. (a) Pipeline: the TMM-pretrained checkpoint regenerated with the recipe of [1] is strict-loaded and fine-tuned (n_train 350 per structure; training seeds 42/123/777); 20 held-out RCWA targets per structure are inverted by Adam on the normalized geometry u with 8 restarts (500 + 300 iterations); the restart with the lowest final surrogate loss is committed; the committed geometry is evaluated once by the RCWA reference solver at its per-wavelength adaptive order; Tier 1A/1B/1C and the T1–T4 taxonomy are accounted. (b) Pre-specified thresholds: $\tau = 5\%$, $k = 2$ (Δ -flag threshold = $k \times$ forward MAE), T1 factor 3, T2 $\varepsilon = 0.05$, T3 factor 0.5, T4 $K = 100$ perturbations with $\sigma = 1\%$ of the box width and a 5 % mean-MAE flag.

Notation. *Structure* denotes one of the absorber families A, B and C of the protocol, or the fourth family D of Section 3.12. *Surrogate MAE* is the mean absolute error of the surrogate's predicted spectrum against the target, and *reference-solver MAE* the same quantity computed from the RCWA solver at the same geometry, both over the 100 wavelength channels and in percentage points. A design is *surrogate-claimed* when its surrogate MAE is at or below τ and *oracle-confirmed* when its reference-solver MAE is; a **pretender** is surrogate-claimed but not oracle-confirmed. Δ is the reference-solver MAE minus the surrogate MAE at the committed geometry, and a design is Δ -flagged when Δ exceeds k times the surrogate's forward test error. $\rho(\text{RCWA MAE}, \text{Surr MAE})$ is the Spearman correlation of the two errors across the N targets of one run. The *committed geometry* $g(u^*)$ is the geometry selected by the commitment rule. *Preliminary r* and *published r* denote the two transferability conventions of Section 2.1.

2.4 Fake-optimum taxonomy

2.5 Pre-specified thresholds and statistical plan

Thresholds: $\tau = 5\%$ (engineering tolerance, the scale of the forward MAE); $k = 2$ (a doubling buffer above the forward noise floor); T1 factor 3; T2 $\varepsilon = 0.05$ (line-edge fabrication scale); T3 factor 0.5; T4

Table 3: Fake-optimum taxonomy (the four types and the diagnostic each one is measured by; nulls are defined in Section 2.5).

Type	Description	Diagnostic
T1 Oracle divergence	RCWA disagrees by more than a factor of three	$\text{MAE}(A_{\text{RCWA}}, A^*) > 3 \cdot \text{MAE}(f_{NN}, A^*)$
T2 Box-edge geometry	recovered geometry pressed to the design box	any normalized parameter within 0.05 of 0 or 1; vs d -dim uniform null
T3 Mode collapse	distinct targets funnel to one region	NN distance $< 0.5 \times$ median pair distance; vs $N = 20$ uniform null
T4 Surrogate-local smoothness	surrogate spectrum under $K = 100$ Gaussian perturbations, $\sigma = 1\%$ box width	mean perturbation MAE $> 5\%$ (measured, not merely defined)

threshold 5 %.

The pre-specified analysis plan:

1. **Primary discrimination statistic:** $\rho(\text{RCWA MAE}, \text{Surr MAE})$ (Spearman, asymptotic p from SciPy [43], Bonett–Wright Fisher-z 95 % CI [44]; no exact or permutation inference is claimed). This is the clean "does surrogate confidence track the reference solver" quantity that H1c tests.
2. $\rho(\Delta, \text{Surr MAE})$ is reported **only as secondary with a coupling caveat:** Δ contains $-\text{Surr}$ by construction, so its negative bias is partly tautological. No "confidence inversion" language is used unless an effect survives in $\rho(\text{RCWA}, \text{Surr})$ itself — which it does not.
3. Cross-structure count comparisons use **Fisher exact** tests [45]; we state outright when differences at $N = 20$ are not significant.
4. T3 clustering can shrink the effective N ; the cluster structure is reported alongside every count-based CI.
5. All seeds are fixed and logged: split 42, train-subset 42, targets 42, restarts `rng(1000 + original-dataset index of the target)`, training 42, with the multi-seed extension at 123 and 777.

Test families and what is confirmatory. The protocol names three confirmatory comparisons — the H1a–c orderings at seed 42 — and does not specify a family-wise correction. Everything else in Section 3 is exploratory and is reported with nominal p -values, with the following corrections applied as sensitivity analyses rather than as decision rules: Holm over the nine per-run Spearman tests and over the nine pairwise Fisher tests of Section 3.4; Holm over the two pooled control primaries of Sections 3.6 and 3.8, and over all ten control tests (six per-structure and two pooled Wilcoxon, two McNemar); and for the 36 detector $\times$ structure intervals of Table 10, an interval excluding 0.5 is read as a selection from that screen, not as a pointwise confirmation. Where several seeds share targets, the shared-target dependence is handled by one of three distinct procedures, not interchangeably: a cluster bootstrap over targets (Table 10; the percentile intervals of Section 3.4 and its 20→50 extension); dividing each structure's counts and denominators by its design effect before the test (the pooled trend of Section 3.3; the effective- N Wilson interval of Section 3.4); or a plain nominal Wilson interval, labelled as such, when neither adjustment is applied.

2.6 Pilot-study defects corrected in the final protocol

Two pilot studies preceded the protocol. A full AI-assisted internal audit of their code found the eight defects listed below; every one was fixed before the protocol was frozen, and no result reported in this paper depends on a pilot run. The complete account, including the pilot headlines that were consequently dropped, is given in Supplementary Section S5.

Table 4: Pilot-study defects and the corrections carried into the final protocol (the complete account is Supplementary Section S5).

#	pilot defect	corrected protocol
1	B/C inverse run on 380–780 nm against 400–1800 nm data	per-structure native grids, pinned in artifacts
2	Adam on raw nm — optimizer frozen within ± 22 nm	normalized variables + multi-start (§2.2)
3	divergent RCWA rows leaked into train/test/targets — an instance of the leakage failure catalogued across ML-based science [37]	Paper 1 filter (§2.1)
4	permuted pretrain input columns; phys stats recomputed on RCWA; C-bounds mismatch	faithful strict loading (§2.1)
5	T4 reported but never computed	implemented and measured (§2.4)
6	$\rho(\Delta, \text{Surr})$ "confidence inversion" headline (coupling artifact)	primary stat = $\rho(\text{RCWA}, \text{Surr})$ (§2.5)
7	A used 6 curated targets vs B's 20 random — the weak-baseline and reporting-bias pattern documented for ML applied to PDE problems [38]	identical $N = 20$ random targets for all structures
8	single deterministic init; <code>manual_seed</code> no-op	$R = 8$ multi-start, archived endpoints

2.7 Pre-specification record

The protocol was frozen on **2026-06-10**, before the seed-42 runs of the as-submitted arm (Section 3.10) and three months before the published-pipeline runs that form the main text, and is reproduced verbatim in Supplementary Section S4. Its SHA-256 is `2e973ba41a82346f4724ff57aaf941dc79534a6e3d485c64dc6c61dd8c559e05` and its git blob id is `e6d020fc3b69011fed4fc39834712da09621b0a9`.

We state the timeline in full rather than claiming more than the record supports. The seed-42 artifacts carry file timestamps of 2026-06-10 14:38–17:59 UTC, consistent with the stated freeze. The protocol file itself was last modified on 2026-06-11 09:39 UTC, and two items in the file as it now stands post-date the freeze. One is the restart-seeding sentence, appended after an AI-assisted code review pointed out that the as-run seeding was `rng(1000 + original-dataset index of the target)` rather than the wording first used; it records the as-run definition, which is a fixed, pre-data quantity, and all results use it. The other is the parenthesis in §5 item 5 naming training seeds 123 and 777 as a multi-seed extension, which was appended after the seed-42 runs. The seed-123 and seed-777 artifacts carry timestamps of 2026-06-10 23:46 to 2026-06-11 07:51 UTC, earlier than the file's last modification, so the record cannot show that the extension was written down before those runs, and the extension is classified as such in Table 5 rather than as pre-specified. What does **not** exist is a contemporaneous public timestamp: the protocol was not committed to version control or deposited anywhere before execution. The surviving protocol is reproduced in Supplementary S4 as preserved, with its two post-freeze additions marked; the repository commit that contains it post-dates the runs. Readers should weigh the pre-specification accordingly: the freeze is documented internally and the analysis plan visibly constrains what we report, but it is not third-party verifiable for this paper.

Amendments made in September 2026, and what was known when. The audit was designed against the release of [1] that was current when the protocol was frozen. Three changes followed, each recorded in the release archive with its date:

1. **2026-09-06 — the audited release changed.** On finding that the companion's printed pipeline differs from the release the protocol was written against (regenerated pools, per-wavelength adaptive Fourier order, complex64, Johnson–Christy materials, and for Structure C a corrected normalization box and an 18-feature set), the whole protocol was re-run on the printed pipeline, which became the main text; the original arm is reported beside it in Section 3.10. Thresholds, hypotheses, target rule and metric hierarchy were not touched.
2. **2026-09-07 — the control experiments were reduced to the canonical seed.** After the printed-

pipeline rates were known (6 of 60 pretenders at seed 42), the from-scratch and feasibility controls were run at seed 42 only rather than at three seeds. The rationale recorded at the time was statistical — that at this base rate three seeds would not give the binary comparison power. An AI-assisted review two days later judged that argument too strong: the seeds share one target set (Section 3.4), so they add correlated replicates rather than independent designs, but replicates do add some precision. The decision was kept on the ground that does hold, cost, and both the original and the revised reasoning are in the release archive. It was in any case a scope decision informed by the main-arm rate, and is disclosed as such.

3. **2026-09-08 — the controls' primary outcome was changed, before any control output existed.** The protocol's endpoint for the controls was the pretender count. At a base rate near 10 % that comparison cannot resolve anything smaller than a rise to 23–35 % (the range covers the possible joint distributions of the paired conditions), so the primary outcome became the **paired shift in Δ** , tested by Wilcoxon signed-rank with a bootstrap interval on the median, with the pretender count retained as a secondary exact-McNemar comparison **and reported whatever it shows**, together with the minimum detectable effect. Those four elements — pairing, the continuous primary, the binary secondary, and MDE reporting — are what the dated amendment fixed. It did not specify how incomplete pairs are handled or a multiplicity correction across the two controls; the analysis script drops a pair whose solver call failed on either side, and the two control comparisons are reported at their nominal p with Holm corrections over the two primaries and over all ten control tests as sensitivity analyses (Section 2.5), both choices made when the script was written and stated here rather than claimed as pre-specified. What was already known was the main arm's rates; what did not yet exist was any control run — the archived analysis file of that date records all six control conditions as absent. This is therefore an amendment informed by the main-arm results and fixed before the control results, not a pre-data plan, and not a choice made after seeing the comparison it governs.
4. **2026-09-16 to 09-19 — the two controls were extended to seeds 123 and 777 after all.** Amendment 2 recorded the decision to run them at seed 42 only. After the seed-42 results and their analysis were written, an AI-assisted review recommended the extension as the more informative of two optional computations, on the ground amendment 2 already conceded — correlated replicates still add precision. The runs used the identical pipeline and the identical analysis script, each seed analysed exactly as seed 42 was, plus a pooled block whose interval resamples targets with their three seeds together (Section 3.4). The extension is post hoc and is reported in Sections 3.6 and 3.8 as a replication of the seed-42 analysis, which is unchanged and remains the pre-specified comparison; the pooled signed-rank and McNemar p -values are labelled nominal because the 179 pairs are $60 \text{ targets} \times 3 \text{ correlated seeds}$.

Table 5 states, for every analysis in Section 3, whether it was pre-specified in that protocol or added afterwards.

3 Results

Every table and figure is regenerated deterministically from the archived per-run artifacts (SHA-256 manifest; Supplementary Sections S1, S2 and S6). Re-executing the GPU training stages reproduces the results statistically rather than bitwise, since CUDA training is not deterministic; the reference-solver stage, by contrast, is bitwise reproducible: re-simulating 12 committed geometries (four per structure) with the cache bypassed, against the vendored upstream commit, returns the archived absorptance spectra exactly on both arms — a maximum absolute deviation of 0 in every spectrum and of 0 percentage points in the resulting MAE (`repro_check_v8.json` in each arm's archive; the published-pipeline recheck at adaptive order took 6.2 GPU-hours). Every archived spectrum also carries a fingerprint of the solver

Table 5: Pre-specification status of every analysis in Section 3 (pre-specified, extension, or post hoc exploratory, with the protocol section that fixes each pre-specified item).

analysis	status	protocol section
H1a–c orderings, seed-42 Fisher and Spearman tests (§3.3)	pre-specified	§1, §5
Tier 1A–1C accounting and the T1–T4 taxonomy (§3.2)	pre-specified	§4
Order-7 spot check of the three highest- Δ geometries	pre-specified, superseded (§3.5)	§3.5
Seeds 123 and 777 (§3.4)	extension; timing relative to the runs not established	§5 item 5
Full order-7 revalidation of all seed-42 designs (legacy arm only)	post hoc exploratory	—
Pooled counts with Wilson intervals, ICC and design effects (§3.4)	post hoc exploratory	—
Re-analysis under the published r and the amplitude ordering (§3.3)	post hoc exploratory	—
Random-search and single-start commitment rules (§3.6)	post hoc exploratory	—
From-scratch (M0) and feasibility-constrained controls, paired Δ primary (§3.6, §3.8)	amended 2026-09-08 after the main-arm rates, before any control output (§2.7)	—
T2-versus-pretender association, CMH (§3.7)	post hoc exploratory	—
Holm correction over the nine tests (§3.3, §3.4)	post hoc exploratory	—
τ sweep and the severity bands (§3.4)	post hoc exploratory	—
Power and minimum-detectable-effect analysis (§3.3)	post hoc exploratory	—
Surrogate-side detector benchmark (§3.7)	post hoc exploratory	—
Geometric feasibility cross-tabulation (§3.8)	post hoc exploratory	—
Lookup null and recoverability (§3.9)	post hoc exploratory	—
Structure D out-of-sample band (§3.12)	recorded before the D runs; amended while 42/50 oracle results existed on disk, uninspected	—
Fixed-order-5 cross-solver probe (§3.11)	post hoc exploratory	—
20→50 target-set extension (§3.4)	post hoc, unregistered; nested (the 20 protocol targets plus 30 more, same trained surrogates), not an independent sample	—

code and settings that produced it and is re-validated against the live solver identity on every read. The multi-seed runs are reported in Section 3.4 and the orders the solver selected in Section 3.5.

3.1 Forward fine-tune fidelity against the published tables of [1]

The fine-tuned surrogates reproduce the forward performance that [1] prints, confirming faithful loading: test absorptance MAE (A for Structures A/B; the mean of A_TE and A_TM for C) is **B 1.72 %, A 1.76 %, C 2.11 %** at seed 42, against the printed ten-seed values of 1.65 ± 0.04 , 1.79 ± 0.04 and 2.05 ± 0.09 % ($|z| \leq 1.7$ on every structure; `pub_gate_table123.json`). The pilot value of 5.17 % arose from unfaithful checkpoint loading, and the as-submitted release audited in Section 3.10 reaches only 3.2–4.8 % because of its narrower bands, looser hygiene and the Structure-C bound mismatch (Section 2.1). These forward MAEs set the per-structure Δ -flag thresholds of $k = 2 \times \text{MAE}$: **3.44 / 3.51 / 4.21 % for B / A / C**.

3.2 Per-structure reliability (seed 42)

Every valid committed geometry was revalidated by the RCWA reference solver (`torcwa 0.1.4.2` [46], 64×64 grid, per-wavelength adaptive Fourier order $N \in \{9, 13, 17\}$ = data-generating fidelity; Section 3.5). The cross-structure picture, structures ordered by the published r ($B > A > C$; Figure 2):

Table 6: Per-structure reliability at training seed 42 (Tier 1A–1C accounting, taxonomy counts and the primary discrimination statistic; columns ordered by the published r of [1], Table 5; preliminary r from [1], Supplementary S19).

Quantity	B ($r = +0.96$; prelim. -0.07)	A ($r = +0.83$; prelim. $+0.72$)	C ($r = +0.65$; prelim. $+0.34$)
n_train	350	350	350
Forward test MAE	1.72 %	1.76 %	2.11 %
Surrogate-claimed success ($\tau = 5$ %)	20/20	20/20	20/20
Oracle-confirmed success	17/20	19/20	18/20
Pretenders [Wilson 95 % CI]	3/20 [5.2, 36.0]	1/20 [0.9, 23.6]	2/20 [2.8, 30.1]
Δ -flagged ($k = 2$)	7/20	2/20	1/20
Δ -flag threshold	3.44 %	3.51 %	4.21 %
Max Δ	12.11 %	9.90 %	4.66 %
Max amplification	35.45×	43.62×	3.25×
$\rho(\text{RCWA, Surr})$ — primary	+0.12 ($p = 0.62$)	+0.34 ($p = 0.14$)	+0.75 ($p < 0.001$)
T1 / T2 / T3 / T4 (of 20)	17 / 15 / 7 / 0	16 / 14 / 0 / 0	1 / 9 / 5 / 0
Distinct modes (of 20)	16	20	17

The two success rows carry the result: **every one of the 20 committed designs passes the surrogate's own success check on all three structures, and the reference solver still rejects 1 to 3 of them per structure**. The threshold test is the part that screens nothing out here: an optimizer that drives the surrogate's own error to 0.06–4.6 % leaves every design below τ , so the verdict cannot separate the 54 the solver confirms from the 6 it rejects. That is an observed property of this optimizer on these targets, not a theorem — a weaker surrogate or a harder target set could fail its own check. That is a weaker statement than the one the same protocol supports on the as-submitted release, where the solver confirmed fewer than half (Section 3.10); here it agrees 85–95 % of the time.

Two rows move in the opposite direction to the pretender counts and are worth reading together with them. B carries **more** Δ -flags than in the as-submitted release (7 of 20 against 6) even though it has a third as many pretenders, because the flag threshold is $k = 2$ times each surrogate's own forward MAE and that MAE has halved (3.44 % here against 6.37 %); Section 4.5 returns to this. And C's primary discrimination statistic is $\rho = +0.75$ ($p < 0.001$), so on that structure the surrogate's continuous self-report does order the reference-solver error even though its pass/fail verdict does not.

3.3 Pre-specified hypothesis tests

The three pre-specified orderings are evaluated under both r conventions in Table 7. **The pre-specified convention is the one H1a holds under; H1b and H1c hold under neither**. H1a — oracle success non-increasing as r falls — is monotone along the preliminary order A, C, B at seed 42 (19 / 18 / 17) and over the pooled runs (57/59, 55/60, 51/60); under the published r it is not. The direction is the one the

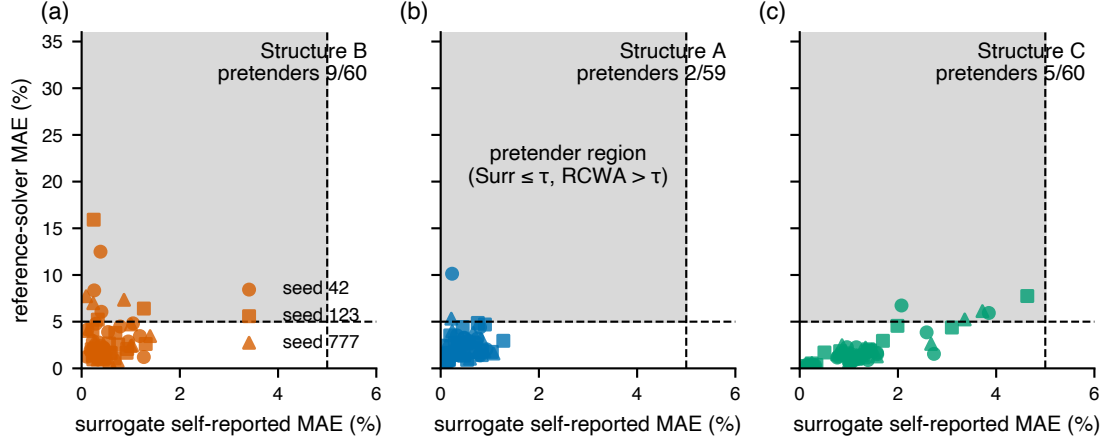


Figure 2: Self-reported versus reference-solver error at every valid committed geometry, for all nine runs (three structures $\times$ training seeds 42, 123 and 777; circles, squares and triangles respectively). Both axes are the mean absolute error (MAE) of the absorptance spectrum against the target over the 100 wavelength channels, in percentage points: on the abscissa as the surrogate reports it at its own committed geometry, on the ordinate as the RCWA reference solver measures it at the same geometry. Dashed lines mark the success threshold $\tau = 5\%$; the shaded upper-left quadrant is the *pretender* region, where the surrogate MAE is at or below τ and the reference-solver MAE is above it. Every point lies left of the vertical line — the surrogate certifies every design — while 16 of the 179 lie above the horizontal one. The counts printed in each panel pool the three seeds.

protocol predicted, but at these counts a pairwise test cannot resolve it: the extreme pair A-vs-B gives Fisher $p = 0.605$.

Table 7: Pre-specified orderings under the two r conventions (seed-42 counts, pooled in parentheses; "monotone" = non-increasing for H1a/H1c, non-decreasing for H1b, along the stated order).

Hypothesis	Quantity (seed 42; pooled)	Monotone under preliminary r (A, C, B)?	Monotone under published r (B, A, C)?	Smallest pairwise p (nominal; Holm)
H1a	oracle success 19 / 18 / 17 (57/59, 55/60, 51/60)	yes	no	0.605 (1.000)
H1b	Δ -flag rate 2 / 1 / 7 (7/59, 1/60, 16/60)	no	no	0.044 (0.392)
H1c	$\rho(\text{RCWA}, \text{Surr}) +0.34 / +0.75 / +0.12$	no	no	—
—	worst-case severity max Δ 9.9 / 4.7 / 12.1 %	no	no	—

A Cochran–Armitage trend test on the pooled pretender counts, scored by the r values themselves rather than by rank, is **nominally significant under the pre-specified convention** ($p = 0.026$) and remains so after the design effect of the shared targets is applied ($p = 0.030$, Section 3.4 — an approximation that divides each structure's counts and denominators by its design effect before the trend test, not a cluster permutation test); under the published r it is not (0.269 nominal, 0.221 corrected) and under the amplitude ordering it is flat (0.981 nominal). Restricted to the canonical seed the same test gives $p = 0.292$, so the trend rests on the pooled runs, which Section 2.7 classifies as exploratory.

Two things bound how much this supports the diagnostic. The trend is carried by **16 events across three structures**, and three points admit a monotone ordering by chance with probability $1/6$ under any single convention — we tested three. And the pairwise tests that would localise the effect do not reach significance. The honest reading is that pretender incidence is **consistent with** the pre-specified ordering in these three structures, not that the ordering is established.

Ordering under the amplitude component. The printed study reads r together with the operating-

band TMM MAE and states that the forward benefit follows the amplitude ordering (A 7.9 % < B 8.9 % < C 16.9 %), not r . Ranked that way the pooled pretender counts are A 2/59, B 9/60, C 5/60 — not monotone, and the trend test is flat ($p = 0.981$). On the published pipeline it is therefore the *preliminary* shape correlation, not the companion's own preferred joint reading, that tracks inverse reliability; on the as-submitted release neither did (Supplementary Section S8).

3.4 Multi-seed robustness

Pooling the three seeds (42 / 123 / 777) tests whether the seed-42 result is specific to that seed:

Table 8: Reliability pooled over the three training seeds (42 / 123 / 777; Wilson intervals on the pretender rates; r column headings as in Table 6, from [1]).

Quantity	B ($r = +0.96$; prelim. −0.07)	A ($r = +0.83$; prelim. +0.72)	C ($r = +0.65$; prelim. +0.34)	Pooled
Surrogate-claimed success	60/60	59/59 [†]	60/60	179/179
Oracle-confirmed success	51/60	57/59	55/60	163/179
Pretenders [Wilson 95 % CI]	9/60 [8.1, 26.1]	2/59 [0.9, 11.5]	5/60 [3.6, 18.1]	16/179 (8.9 %) [5.6, 14.0]
Δ -flagged ($k = 2$)	16/60	7/59	1/60	24/179
ρ (RCWA, Surr) per seed	+0.117 / +0.146 / −0.119	+0.340 / +0.454 / +0.186	+0.750 / +0.829 / +0.719	range −0.119 . . . +0.829

[†]One A geometry at seed 777 was degenerate and its solver call failed, so A's denominator is 59 (179 valid of 180). In the as-submitted release the failing geometry was in C instead.

The pooled rate is **16/179 = 8.9 %** (per-run counts and intervals in Supplementary Table S1; Figure 3a). Its nominal Wilson 95 % CI is [5.6, 14.0] %, but the 20 targets of a structure are bit-identical across its three seeds, so the runs are correlated: the intraclass correlation is 0.79 on C, 0.10 on B and −0.02 on A, for design effects of 2.58 / 1.21 / 0.96 and effective sizes of 23 / 50 / 61. The pooled design effect of 1.33 leaves the information of ~134 designs, widening the interval to [5.2, 15.0] % (cluster bootstrap: [4.0, 14.5] %), so we treat the per-seed counts (B 3/3/3, A 1/0/1, C 2/1/2 of 20; A at seed 777 of 19) as primary. The rate is strongly τ -dependent (Figure 4), and this is the most consequential number in the section: tightening the tolerance to $\tau = 2.5$ % turns **53 of the 171 designs the surrogate still certifies into pretenders (31 %)**, while loosening it to 7.5 % leaves 6/179 and to 10 % leaves 3/179. Of the $\tau = 5$ % pretenders 10 are mild (5–7.5 %), 3 moderate and 3 severe (≥ 10 %). The pretender rate is therefore a property of the tolerance the design task demands, not a fixed property of the surrogate.

As a post-hoc target-set extension — not a redefinition of the primary 20-target protocol above, and not an independent replication of it — the same trained surrogates were run on 50 targets per structure, the original 20 plus 30 more drawn by the identical mechanism; the 20-target counts of Table 8 are therefore a subset of the 50-target counts below, not a separate sample. Pooling the three seeds over $50 \times 3 = 150$ target-seed slots per structure (147 valid on A, where three slots hit the same class of degenerate-geometry solver failure as in the 20-target run; 150 on B and C) gives a pooled pretender rate of **42/447 = 9.4 %** (nominal Wilson 95 % CI [7.0, 12.5]; design-effect-adjusted [6.8, 12.9] at $n_{\text{eff}} = 348$; cluster bootstrap over targets, 20,000 reps, [6.3, 12.8]) — overlapping the 20-target estimate of 8.9 % [5.2, 15.0]. The per-structure ordering is unchanged (A 5/147 = 3.4 %, C 14/150 = 9.3 %, B 23/150 = 15.3 %), the per-structure design effects are similar in direction and magnitude (A 0.94, B 1.17, C 2.38 at 50 targets vs. 0.96 / 1.21 / 2.58 at 20 targets), and the τ -sensitivity pattern is repeated on the larger set: tightening τ to 2.5 % turns 155 of the 426 still-certified designs into pretenders (36.4 %, cluster bootstrap [30.9, 42.2]) at 50 targets, the same qualitative collapse as the 31 % figure at 20 targets. Because the two counts overlap and share the same trained surrogates, this is descriptive consistency with the original estimate, not an independent confirmation of it; we therefore continue to treat the 20-target result as the pre-specified

primary estimate and report the 50-target numbers as an unregistered sensitivity check that does not, on its own, change the sign or approximate magnitude of that estimate.

Whether the continuous self-report carries rank information depends on the structure. $\rho(\text{RCWA}, \text{Surr})$ spans -0.119 to $+0.829$ over the nine runs; four are nominally significant — all three C runs ($+0.750, +0.829, +0.719$; $p \leq 0.0004$) and A s123 ($+0.454, p = 0.044$) — and **the three C runs survive the nine-test Holm correction** (adjusted $p = 0.0011, 0.00006, 0.0025$) while A s123 does not (0.27). (Exploratory) seed-pooled block permutation gives $\rho = -0.01$ (B, $p = 0.94$), $+0.33$ (A, $p = 0.080$), $+0.76$ (C, $p = 0.0002$). As a detector of pretenders the surrogate's own claimed MAE has a pooled AUROC (Table 10's seed-stratified estimator) of 0.40 on B, 0.27 on A and **0.98 on C**: on B and A the pretenders are, if anything, the designs the surrogate is *most* confident about, whereas on C the surrogate's error ranking is nearly perfect even though its pass/fail verdict admits every design (Section 3.7).

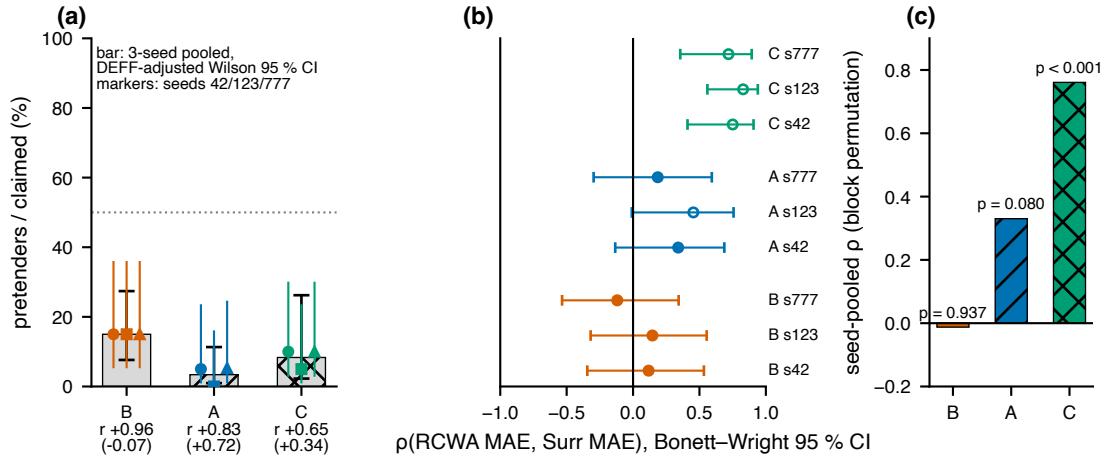


Figure 3: Transferability versus inverse reliability. A design is *surrogate-claimed* when the surrogate's own mean absolute error (MAE) against the target is at or below $\tau = 5\%$, and a *pretender* when it is claimed but the RCWA reference solver puts the MAE above τ . (a) Pretenders per surrogate-claimed design for each run (markers: training seeds 42, 123 and 777, with Wilson 95% intervals) and the three-seed pooled rate (bar) with a design-effect-adjusted Wilson interval; dotted line: coin flip. Tick labels give the published r and, in parentheses, the preliminary r in force when the protocol was frozen — both are measurements of [1] (Table 5; Supplementary S19); the inverse-reliability quantities are from the present study. (b) $\rho(\text{reference-solver MAE}, \text{surrogate MAE})$ per run with Bonett-Wright 95% intervals; open markers are the four nominally significant runs. (c) Seed-pooled within-structure ρ with block-permutation p (targets permuted jointly across seeds; exploratory).

3.5 Per-wavelength adaptive order of the reference solver

The published pipeline chooses the Fourier truncation per wavelength from $N \in \{9, 13, 17\}$ by the companion's feature-size rule rather than fixing it, so the fidelity of the audit varies with the geometry it is asked to resolve rather than with a number set in advance. The audit therefore evaluates every committed geometry at exactly the fidelity its training labels were generated at — the same vendored routine decides the order — which is what "reference fidelity" means throughout. Across the 18 000 wavelength solves behind the nine runs of Sections 3.2 and 3.4 the solver used $N = 17$ on **59%** of them, $N = 13$ on 7% and $N = 9$ on 35% (`adaptive_orders_v10.json`), and the split is strongly per structure: Structure A runs at $N = 17$ on 98% of its wavelengths and C on 75%, while B is at $N = 9$ on 83%. That is also where the cost goes: a committed A geometry takes about 58 min to validate against 3 min for B. That cost is the one the companion's convergence study reports per wavelength at $N = 17$ (≈ 28 s; [1], Supplementary S16) multiplied by the 100-wavelength grid; nearly every A geometry, in the dataset as in the committed set, has a metal feature or gap below the rule's 150-nm threshold and is therefore solved at $N = 17$ throughout.

This replaces the fixed-order-7 revalidation the protocol specified. That check existed to bound the error of a solver pinned at order 5 — the companion's convergence study ([1], Supplementary S16) reports

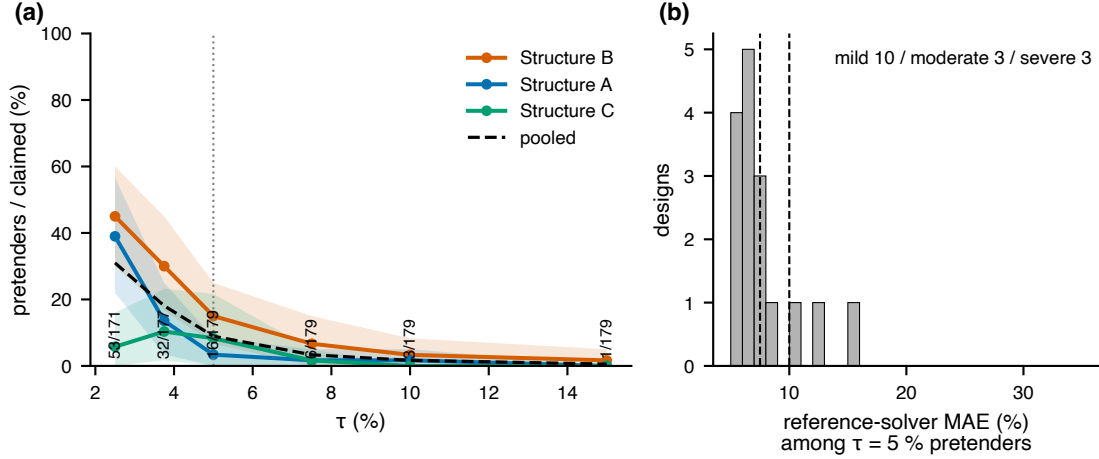


Figure 4: Sensitivity to the success threshold (exploratory; all nine runs, training seeds 42, 123 and 777 pooled). (a) Pretenders per surrogate-claimed design as τ is varied, per structure and pooled, where at each τ a design counts as claimed if the surrogate’s mean absolute error (MAE) is at or below τ and as a pretender if the reference-solver MAE additionally exceeds τ ; bands are a cluster bootstrap over the 20 targets of each structure and the labels give the pooled pretender/claimed counts, whose denominator falls from 179 to 171 as τ tightens to 2.5 %. (b) Distribution of the reference-solver MAE among the $\tau = 5$ % pretenders, with the post-hoc mild / moderate / severe band edges at 7.5 % and 10 %.

$N = 5$ absorbance discrepancies of ≈ 2 pp on A, ≈ 3 pp on B and ≈ 5 pp on C (TE), reaching ≈ 6 pp and ≈ 15 pp respectively at long wavelengths — and re-running at 7 would now *lower* the fidelity of a solver already at 9 to 17. The order-7 results for the as-submitted arm, where they do apply, are in Supplementary Section S3. What the adaptive order does **not** give is a convergence proof: the solver reports cases it cannot converge at its cap of 17, so these are reference-fidelity statements (Section 4.7, W8). Section 3.11 re-solves the same committed geometries, on the same 64×64 raster, at a fixed order 5 and measures what the order alone contributes.

3.6 Commitment rules and the from-scratch control (exploratory)

Is the pretender phenomenon a gradient-exploitation artifact, or a generic consequence of committing to the surrogate’s best point? At seed 42 we compare three commitment rules under a matched evaluation budget:

Table 9: Pretenders under three commitment rules at a matched evaluation budget (seed 42; r headings from [1]; pretenders / surrogate-claimed designs, the same quantity Figure 6 plots, with the oracle-confirmed count in parentheses. Supplementary Sections S2 and S8.2 give both quantities for both arms.)

Commitment rule	B ($r = +0.96$; prelim. -0.07)	A ($r = +0.83$; prelim. $+0.72$)	C ($r = +0.65$; prelim. $+0.34$)
Budget-matched random search	6/20 (14 confirmed)	4/20 (16 confirmed)	1/16 (15 confirmed)
Gradient, best of 8 restarts	3/20 (17 confirmed)	1/20 (19 confirmed)	2/20 (18 confirmed)
Gradient, single start	4/19 (15 confirmed)	4/19 (15 confirmed)	4/15 (11 confirmed)

(pretenders over surrogate-claimed designs, so the denominator is the number of designs each rule certified; Figure 6; full table with both denominators, mean oracle MAEs, and Fisher tests in Supplementary Section S2). **Every rule manufactures pretenders**, so the gradient loop is **not necessary** for the phenomenon, in the same sense Section 3.8 uses for transferred weights and geometric infeasibility: removing it does not remove pretenders, but this comparison does not isolate the gradient loop’s own contribution, if any. Budget-matched random search produces more of them than best-of-eight gradient descent on B and A (6/20 against 3/20; 4/20 against 1/20) and fewer on C (1/16 against 2/20), and none of

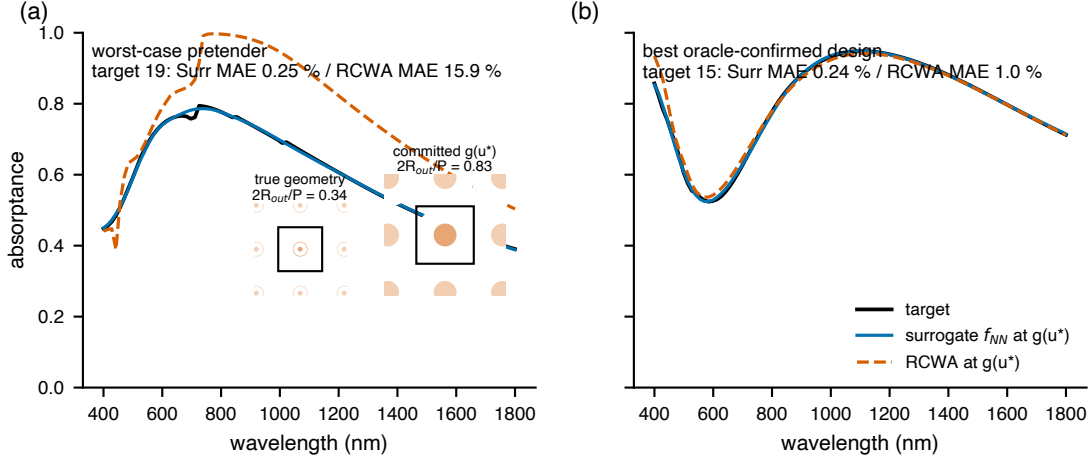


Figure 5: Structure B, seed 123, absorbance channel. (a) Worst-case pretender (target 19 of 20): target spectrum, surrogate prediction at the committed geometry $g(u^*)$ with self-reported MAE 0.25 %, and the reference solver at the same geometry with MAE 15.9 %. The committed ring has $R_{in} = 253$ nm against $R_{out} = 208$ nm — an inner radius larger than the outer one, which the generator’s sampling rule ($R_{in} \leq R_{out} - 10$ nm) can never produce, so the optimizer has placed its optimum outside the generator-feasible region entirely rather than merely at its edge. Its true counterpart is an ordinary ring ($R_{out} = 114$ nm, $R_{in} = 104$ nm, $2R_{out}/P = 0.34$). (b) Best oracle-confirmed design of the same run (target 15: self-reported MAE 0.24 %, reference-solver MAE 0.96 %), whose self-report is indistinguishable from the pretender’s. Target spectra are held-out samples of the published Structure-B dataset of [1]; the surrogate and reference spectra at the committed geometries were computed for this study.

the three comparisons is significant at these counts (Fisher $p = 0.45 / 0.34 / 1.00$). Single-start gradient is at or above best-of-eight everywhere (4/19, 4/19, 4/15): this argues against restart selection pressure being what *creates* pretenders, but — as with the other two controls — it does not by itself identify what does. What the pub-arm counts no longer support is the stronger claim that a weaker optimizer is *reliably* worse: on C random search both certified fewer designs (16 of 20) and had the single lowest pretender count. Selecting the argmin of an imperfect learned objective is enough, on its own, to produce pretenders under every one of the three rules tested — consistent with the surrogate-gaming pathology of offline model-based optimization [13, 14, 15, 16], here on a physical inverse-design testbed — but, as in Section 3.8, this does not establish it as the sole or necessary cause, and this study cannot rank the commitment rules against each other.

From-scratch control. The same architecture trained from random initialization rather than from the TMM-pretrained checkpoint (M0, learning rate 10^{-3} from the upstream from-scratch recipe) reaches a forward test MAE of 1.89, 2.59 and 1.96 % on B, A and C against the transfer-learned 1.72, 1.76 and 2.11 %, so it is a surrogate of comparable accuracy, and it draws the same targets — the split is fixed before the training subset, so M0 and the transfer-learned run are paired design by design. Over the 60 paired designs the signed-rank test does not reject no-shift (Wilcoxon $p = 0.176$); its estimand, the Hodges–Lehmann shift in Δ , is **+0.22 pp** [−0.09, +0.58], the median shift is +0.14 pp [−0.07, +0.44] (bootstrap), the mean +0.06 pp, and 27 designs move down against 33 up. The pretender counts go **6** → **4** (discordant 4/2, exact McNemar $p = 0.688$). Per structure the median shift is B −0.36 pp ($p = 0.216$), A +0.75 pp ($p = 0.0017$; Hodges–Lehmann +0.92 pp [+0.32, +1.67]), C −0.03 pp ($p = 0.622$): the from-scratch A surrogate, whose forward MAE is 47 % higher than the transfer-learned one, is measurably worse at the committed geometries, while B and C are not.

Replication at seeds 123 and 777 (post hoc, Section 2.7 amendment 4). The same control at the other two training seeds gives Hodges–Lehmann shifts of +0.17 pp [−0.15, +0.50] (Wilcoxon $p = 0.283$; pretenders 4 → 6) and +0.42 pp [+0.15, +0.73] ($p = 0.0025$; 6 → 7), and the from-scratch A surrogate is again the less accurate one at every seed (forward MAE 2.62 and 2.57 % against 1.78 and 1.68 %, with B and C within 0.2 pp of their transfer-learned counterparts). Over the 179 pairs of all three seeds the median

shift is **+0.20 pp** [**+0.03, +0.38**] with the interval clustered by target (a target's three seeds resampled together; Hodges–Lehmann +0.27 pp, nominal), the pretender counts are **16** → **17** (discordant 11/12), and the per-structure pattern replicates: A +0.82 pp [+0.55, +1.32] with pretenders 2 → 7, B +0.02 pp [−0.24, +0.61] with 9 → 5, C +0.01 pp [−0.10, +0.14] with 5 → 5 (`control_analysis_v10_3seed.json`).

Transferred weights are therefore not necessary for pretenders: a surrogate trained without them produces them at a rate this design cannot distinguish from the transfer-learned one (6 against 4 at the pre-specified seed, 16 against 17 over three), and the pooled shift in Δ is small and positive — about a fifth of a percentage point over three seeds, carried entirely by the structure on which the from-scratch surrogate is less accurate. The interval is what carries that statement — the binary comparison on its own could not distinguish 6 from 4, and at this base rate it would only reach 80 % power against a control rate of 23–35 % depending on how the two conditions co-occur, so we report the bound rather than a bare null. Section 2.7 records that this analysis was fixed after the main-arm rates were known and before any control output existed.

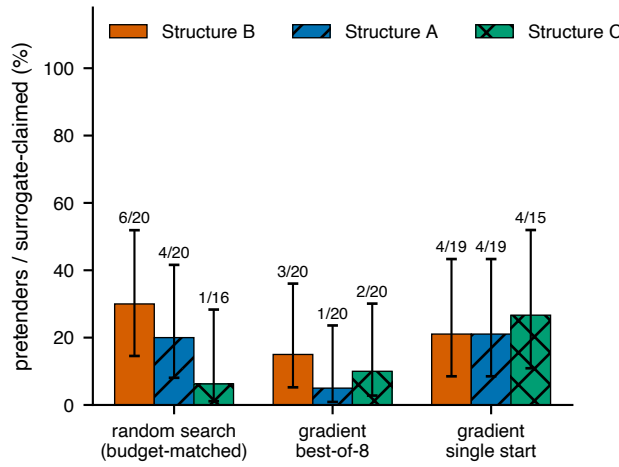


Figure 6: Pretenders per surrogate-claimed design at training seed 42 under three rules for choosing which geometry to commit: random search over 6400 uniform draws from the design box — *budget-matched*, i.e. given the same number of surrogate evaluations as the gradient loop's 8 restarts $\times$ 800 iterations — gradient descent with the best of 8 restarts, and gradient descent from the single mid-box start. Wilson 95 % intervals; k/n labels give pretenders over surrogate-claimed designs, so the denominators differ between rules (C random search: 1/16). Committing to the surrogate argmin suffices to produce pretenders under every rule, without isolating which of the three commitment mechanisms, if any, is the larger contributor; no pairwise difference between the rules is significant at these counts.

3.7 Surrogate-side diagnostics

No cheap, oracle-free diagnostic flags the pretenders on every structure:

- **T4 surrogate-local smoothness** (measured, $K = 100$): **0/179 flags**; the mean perturbation MAE averages 0.79–1.12 % per run and never exceeds 2.97 %. The pre-specified finite-scale perturbation threshold is therefore not exceeded at any committed geometry; this does not rule out roughness at other scales or directions.
- **Multi-start dispersion** (Tier 1C): the spread of the eight restarts' final losses is small (standard deviation 5×10^{-4} to 2.4×10^{-3} in MSE units per run, against committed losses of order 10^{-4} – 10^{-3}) while the *endpoint geometric* spread is large (1.36–1.88 in u-space) — distinct basins with comparably low claimed loss, itself uninformative about which basin is real. The loss-tail RSD of the committed run (0.002–0.012) indicates stable terminal loss traces; it does not establish that the geometry or gradient had stopped changing.
- **T2 box-edge** flags remain abundant (9–18/20 per run) but the association with pretender status that the as-submitted release showed **does not reproduce here**: the Cochran–Mantel–Haenszel estimate

is OR 2.18 with $p = 0.246$, and no per-run Fisher test is significant. As a detector it is worse than before in the way that matters: sensitivity 0.75 and specificity 0.37 give a PPV of 0.10 against a base rate of 0.089, so flagging every box-edge design would reject nearly two thirds of the good designs to catch three quarters of a much smaller bad set. The flag also partly proxies geometric infeasibility (Section 3.8). At seed 42, T3 mode collapse appears on B and C (7 and 5 of 20) but not A; A does exhibit T3 at seed 123, so this is a seed-42 statement.

None of them is usable as a detector on its own across structures. Pooling the three seeds within each structure gives the following areas under the ROC curve for separating pretenders from oracle-confirmed designs (0.5 = chance):

Table 10: Discrimination of the oracle-free diagnostics (area under the ROC curve for separating pretenders from oracle-confirmed designs; the pooled value is the per-run AUROC averaged over the three seeds with weights $n_{\text{pretender}} \times n_{\text{confirmed}}$, so that seeds are never compared across runs, and the interval is a 2000-draw cluster bootstrap over targets in which a target's three seeds are resampled together; draws without a pretender are discarded, so the intervals are conditional on that: on A, with two pretenders, 1716–1779 of the 2000 draws were usable and its intervals are exploratory (C 1744–1780, B 1998–2000); 36 cells are screened, so an interval excluding 0.5 is a selection, not a pointwise confirmation; 0.5 = chance; `detector_bench_v8.json`).

oracle-free quantity	B AUROC [95 % CI]	A AUROC [95 % CI]	C AUROC [95 % CI]
3-seed ensemble disagreement	0.80 [0.63, 0.94]	0.73 [0.42, 1.00]	0.95 [0.88, 1.00]
held-out ensemble MAE	0.75 [0.53, 0.91]	0.51 [0.26, 0.79]	0.98 [0.89, 1.00]
ensemble-mean MAE	0.73 [0.52, 0.89]	0.38 [0.16, 0.60]	0.98 [0.84, 1.00]
distance to the fine-tune set	0.80 [0.63, 0.94]	0.73 [0.50, 0.94]	0.42 [0.16, 0.61]
distance to the dataset pool	0.79 [0.62, 0.92]	0.73 [0.50, 1.00]	0.49 [0.30, 0.78]
TMM-surrogate disagreement	0.42 [0.23, 0.63]	0.62 [0.39, 0.84]	0.23 [0.00, 0.46]
TMM-target disagreement	0.43 [0.24, 0.63]	0.62 [0.38, 0.84]	0.22 [0.00, 0.44]
T4 local-smoothness value	0.48 [0.20, 0.77]	0.92 [0.79, 1.00]	0.66 [0.44, 0.87]
T2 box-edge flag	0.62 [0.49, 0.75]	0.66 [0.57, 0.77]	0.46 [0.35, 0.60]
multi-start final-loss spread (std over the 8 restarts)	0.50 [0.29, 0.68]	0.27 [0.00, 0.63]	0.74 [0.54, 0.92]
the surrogate's own claimed MAE	0.40 [0.19, 0.67]	0.27 [0.09, 0.49]	0.98 [0.89, 1.00]
generator-infeasibility flag	0.76 [0.68, 0.86]	0.74 [0.63, 0.87]	0.37 [0.28, 0.46]

The picture is structure-specific rather than uniformly null. On **C**, every ensemble-based quantity and the surrogate's own claimed MAE reach 0.95–0.98 with intervals excluding 0.5: the C surrogate knows, in rank order, which of its designs are wrong: in every C run the pretenders sit among the four designs with the largest claimed MAE, so re-simulating the worst-ranked fifth of each run would catch all of them. On **B**, the distance to the nearest dataset point (0.79–0.80), the ensemble quantities (0.73–0.80) and the infeasibility flag (0.76) exclude 0.5 but none exceeds 0.81, while the claimed MAE is *below* chance (0.40). On **A**, with only two pretenders in 59 designs, only the T4 perturbation value (0.92), the two binary flags (0.66, 0.74) and the claimed MAE — again below chance (0.27) — have intervals excluding 0.5. No single quantity separates the two classes on all three structures, and the one that works best on C — the surrogate's own error — is the one that ranks pretenders as the *most* trustworthy designs on A and B.

Reference-solver evaluation determines the audit's labels by definition, and no tested diagnostic perfectly separated pretenders from confirmed designs on every structure: one oracle call per design (103–4595 s at the adaptive order, median 1655 s), applying the decision rule of Section 2.3 (surrogate-claimed, $\text{Surr MAE} \leq \tau$, **and** reference-solver $\text{MAE} > \tau$). Δ , the gap between the two MAEs, is not that decision rule — it is reported separately, as severity (Section 2.3, Tier 1B).

3.8 Geometric feasibility of the committed designs (post hoc)

The datasets are generated under structure-specific fabrication constraints ([1], Supplementary S17) — patch widths at most 0.9 P for A and C, and for B an outer ring radius at most 0.45 P with each inner radius at least 10 nm smaller than the one enclosing it — but the optimizer searches the raw sampling box, which does not enforce them. It leaves it: over the three seeds, 33/60 committed B geometries, 33/59 A

and 13/60 C violate a constraint that every training sample satisfies. This is not a surprise — a uniform draw from the box is infeasible with probability 0.77 on B, 0.56 on A and 0.52 on C — but on B it matters, because there every pretender is infeasible: 9 of the 33 infeasible B designs are pretenders against 0 of the 27 feasible ones (Fisher $p = 0.0029$; run-stratified CMH $p = 0.0036$). On A the two pretenders are also infeasible but the count is too small to test (2/33 against 0/26, $p = 0.50$), and on C the association runs the other way (0/13 against 5/47, $p = 0.58$): the C pretenders are generator-feasible. A feasibility check is free, requires no solver call, and on B is the one oracle-free signal that both carries information (Table 10, AUROC 0.76) and needs no ensemble, which is why Section 3.7 is stated per structure rather than as a blanket negative. Figure S1 shows where in the design box the committed geometries sit.

Constrained rerun. Repeating the seed-42 inverse design over a nested parametrization whose image is exactly the generator's feasible region — every candidate generator-feasible by construction, so the optimizer cannot leave it — changes the outcome less than the B result alone would suggest, and the distribution of the change is asymmetric enough that one number does not describe it. Paired over the 60 designs the two runs share, the signed-rank test rejects no-shift ($p = 0.0071$; 40 designs improve, 20 worsen, sign test $p = 0.013$): the Hodges–Lehmann shift is **−0.18 pp** [−0.58, −0.00], the median shift −0.00 pp [−0.21, −0.00] (bootstrap), and the mean −0.79 pp because a few designs improve by several points (range −10.7 to +4.6 pp). Pretenders fall **6** → **3** (discordant 4/1, exact McNemar $p = 0.375$). By structure the median shift is B −0.55 pp ($p = 0.012$; Hodges–Lehmann −1.39 pp [−2.91, −0.16], mean −1.71 pp), A −0.15 pp ($p = 0.154$; mean −0.71 pp), C −0.00 pp ($p = 0.841$; mean +0.04 pp) — the effect is largest on B, the structure that commits the most infeasible geometries and whose pretenders are all infeasible, with A contributing the remainder of the pooled mean. Treated as one family with the other nine control tests, the pooled feasibility p becomes 0.063 after Holm correction (`control_analysis_v10.json`); the two pooled primaries as a family of two give 0.014.

Replication at seeds 123 and 777 (post hoc, Section 2.7 amendment 4). The feasibility-constrained rerun at the other two seeds takes the pretender count 4 → 1 and 6 → 1 (Hodges–Lehmann shifts −0.02 pp [−0.26, +0.00], Wilcoxon $p = 0.298$, and −0.09 pp [−0.51, −0.00], $p = 0.019$). Over all three seeds the count goes **16** → **5** (discordant 12/1; exact McNemar $p = 0.003$, nominal), the median shift stays negligible at −0.00 pp [−0.02, −0.00] (target-clustered) while the mean is −0.60 pp, and the structure pattern is the seed-42 one: B −0.24 pp [−0.89, −0.01] with pretenders 9 → 1, A −0.03 pp [−0.50, +0.01] with 2 → 0, C −0.00 pp with 5 → 4 (`control_analysis_v10_3seed.json`). The five that survive confinement are four C designs and one B design.

Confinement to the generator-feasible region therefore reduces but does not remove the problem: the median shift is negligible, B — the structure whose pretenders are all infeasible — improves the most, and pretenders persist when the optimizer cannot leave the feasible region at all. Together with the from-scratch control of Section 3.6 and the commitment rules of Table 9, the three controls show that **none** of transferred weights, the gradient loop or geometric infeasibility is *necessary* for pretenders: each can be removed and pretenders remain. They do not show that any of the three contributes nothing — the feasible rerun takes B from 3 pretenders to 1 and the pooled count from 6 to 3 — and they do not isolate a single cause. What the three controls share is the one step none of them removes, committing to the argmin of a surrogate that is imperfect where it is most optimistic.

3.9 Realizable targets and recoverability (post hoc)

Two further questions bear on how much the loop is worth. First, could the practitioner have skipped it? Every target is a held-out dataset spectrum, so a nearest-neighbour lookup in the fine-tune set is a legitimate baseline: at seed 42 it already lands within τ for 20/20 targets on B, 19/20 on A and 11/20 on C, with median errors of 2.5, 3.2 and 4.8 %. Against that, the optimizer's oracle-confirmed successes are 17/20, 19/20 and 18/20 (Table 6): on B the lookup does at least as well as the loop under the protocol's threshold, on A the two tie, and on C — where the surrogate ranks its own designs correctly — the loop beats the lookup by seven targets. The optimizer's *claimed* errors (median 0.4, 0.4 and 1.1 %) are far below either. Second, is the surrogate able to recognise the right answer? Evaluated at the true generating geometry it reports a median error of 1.6 % on B, 1.7 % on A and 1.8 % on C — within τ on 20, 20 and

18 of the 20 targets, but worse than what it claims at its own committed point on 100 % of targets on all three structures. The committed geometries are far from the true ones (median $\|u^* - u_{\text{true}}\| = 0.64, 0.78$ and 0.31 in box units), which is expected — these inverse problems admit many solutions by construction, so a different geometry is not itself a failure, and preferring one's own solution to the generating one is not either. The failure is the 16 cases where the preferred solution is wrong.

3.10 The same protocol on two versions of one pipeline

The audit was also run end to end on the release this pipeline superseded — the 2026-03 as-submitted version of [1], with its fixed Fourier order 5, complex128 arithmetic, and the data pools and checkpoints of that release — under the identical protocol, thresholds and target-drawing rule. Nine runs finished in both arms, and the counts are read from `pub_vs_legacy_v10.json`:

Table 11: The identical protocol on the two versions of the pipeline (nine runs per arm; intervals nominal in the sense of Section 3.4).

	as-submitted release	published pipeline
forward test MAE	3.19–4.84 %	1.58–2.11 %
surrogate-claimed ($\tau = 5$ %)	179/179	179/179
oracle-confirmed	84	163
pretenders [Wilson 95 %]	95/179 = 53.1 % [45.8, 60.2]	16/179 = 8.9 % [5.6, 14.0]
per structure	B 30/60 · A 27/60 · C 38/59	B 9/60 · A 2/59 · C 5/60

The difference is 44.1 pp (Newcombe 95 % CI [35.2, 52.1], Fisher exact $p = 2.3 \times 10^{-20}$); it holds in every structure (Fisher $p = 7.4 \times 10^{-5}, 5.8 \times 10^{-8}, 6.9 \times 10^{-11}$ for B, A, C). Resampling targets within each arm and structure, a target's three seeds together, which respects the clustering of Section 3.4, leaves the interval at [33.0, 54.8] pp.

What this does and does not identify. The two arms differ in their checkpoints, their training pools, structure A's wavelength band, structure C's feature set and bounds, and the reference solver's truncation order, precision and material model — all at once, and their target sets differ because the pools do. So this is a comparison of two complete pipelines, not an ablation: it does not attribute the change to any one of those factors, and the intervals above are nominal in the sense of Section 3.4. What it does establish is that **a reliability statement of this kind is a property of a specific release, not of a structure or of a method:** the same protocol, the same three geometries and the same thresholds move the pretender rate by a factor of six between two versions of one pipeline. Section 3.11 separates the solver's contribution from the rest.

3.11 Fixed order 5 on the same committed geometries

The two-arm contrast of Section 3.10 bundles every difference between the releases — data, hygiene, checkpoints, bounds and the reference solver's truncation — so it cannot say how much of the sixfold gap is the solver. This probe isolates that one factor: the 60 seed-42 geometries the published-pipeline surrogates committed were re-solved with the **same** module, precision, materials and grid but the Fourier order fixed at 5, the as-submitted release's setting, instead of chosen adaptively from {9, 13, 17}. Each design is therefore compared with itself on the same raster, and only the order changes (`cross_solver_v11.json`).

Result. At order 5 the reference MAE is higher on 46 of the 60 designs and lower on 14 (median shift +0.61 pp, median $|\text{shift}|$ 0.90 pp, Wilcoxon $p = 1.4 \times 10^{-4}$), with design-level excursions of up to 4.2, 4.4 and 6.1 pp on B, A and C. Those excursions sit where they matter: the pretender count on the identical designs goes **6 → 13 of 60** (B 3 → 6, A 1 → 4, C 2 → 3), every flip in the same direction — seven designs the adaptive solver confirms are rejected at order 5, none the other way (exact McNemar p

= 0.016). The two solvers agree on the ranking only moderately (Pearson $r = 0.74, 0.73, 0.56$), and the order-5 solve is 10–120 times faster (16–29 s per design against 156–3526 s).

What it establishes, and what it does not. On the same surrogates and the same committed geometries — the published pipeline's — truncating the reference solver to order 5 alone doubles the pretender count from 6 to 13 of 60. That is a clean, controlled result: a reference solver truncated at order 5 rejects designs that the same solver at its adaptive order accepts. It does **not** decompose the as-submitted arm's own seed-42 count of 31 of 60 (Supplementary Section S8): that arm's committed geometries are different designs, generated by different checkpoints and data pools, so this probe was never run on them, and how much of its 31 owes to truncation versus to the narrower bands, looser hygiene, the Structure-C bound mismatch or the surrogates trained on them is not established by this or any other analysis in this paper. What the probe does support for reading Section 3.10 is that the published pipeline's own reported rate is itself sensitive to the audit's choice of reference fidelity — order 5 would have reported 13 rather than 6 on identical designs — so the 8.9 % is not immune to the same kind of truncation dependence, whatever the as-submitted arm's own truncation dependence turns out to be. Which verdict is physically right is not established by either — the adaptive solver is the reference fidelity of Section 3.5, not a convergence proof. What the higher order does not change is the rest: 6 of these 60 designs fail at the reference fidelity, and no truncation setting makes the surrogate's self-check informative.

3.12 Out-of-sample test on a fourth structure

The three structures of the protocol were chosen because the pilot r ordered them; a fourth structure that did not exist when the protocol was frozen is a cleaner test of whether the diagnostic says anything about inverse reliability, because nothing about the audit was tuned on it. Structure D is a cross-shaped Cr patch (four-fold symmetric and polarization-independent at normal incidence; six parameters $P, L, w, t_{\text{Cr}}, d_{\text{SiO}_2}, \theta$; 11 physics features). The present audit uses its TE response at oblique incidence. It was generated on 2026-07-08 as a contingency structure for the companion's re-review with the same published-pipeline settings as the main arm (400–1800 nm, adaptive order, complex64, Johnson–Christy; 494 of 500 samples reliable; dataset beb3c107... , regenerated TMM checkpoint 4ab5babf...). Its pilot diagnostic, computed with the companion's printed estimator, is $r = +0.905$ with an operating-band TMM MAE of 10.51 % — between B and A on r , between B and C on amplitude — and the fine-tuned surrogate reaches a forward test MAE of 1.30 %, the lowest of the four. The audit ran on the full 50-design held-out split at seed 42.

The recorded prediction. Before any D inverse design or oracle call, we wrote down (2026-09-10, `structure_D_prespecification.md` in the repository) that if the diagnostic carries information, D's pretender rate should fall between its neighbours', **5–15 %, i.e. 3 to 7 of 50**, and that a rate below A's 3.4 % or above B's 15.0 % would falsify that reading. The same file records that its arithmetic and one mislabelled bound were corrected on 2026-09-12 while 42 of the 50 oracle results already existed on disk and were not inspected — so this is an amendment made with partial results in existence, not a blinded pre-specification, and it should be read as such.

Result. All 50 committed designs pass the surrogate's own check, and the reference solver rejects **4 of 50 (8.0 %, Wilson [3.2, 18.8])**: inside the recorded band, not falsifying. Two of the four are mild (5.3 and 6.4 %) and two severe (12.3 and 15.2 %); the worst commits an arm width $w = 278$ nm larger than the arm length $L = 181$ nm, a cross that the generator cannot produce. That is the pattern of Section 3.8 again: 20 of the 50 committed D geometries violate the generator's rule ($L \leq 0.9 P$ and $w \leq L$), **all four pretenders are among them, and none of the 30 feasible designs is a pretender** (Fisher $p = 0.021$, post hoc). The Δ -flag fires on 7 of 50 at a threshold of 2.61 % ($k = 2 \times 1.30$ %), the surrogate's continuous self-report is informative here ($\rho = +0.51$, $p = 0.0002$, between A's and C's values), and T4 finds the landscape smooth at every design (0 flags; mean perturbation MAE 0.92 %, maximum 2.27 %). The solver ran at $N = 17$ on 45 of the 50 designs, a median of 1505 s per design.

What it does and does not show. Placed on the four-point ordering by the printed r ($B > D > A > C$) the rates are 15.0, 8.0, 3.4 and 8.3 % — not monotone, Cochran–Armitage $p = 0.32$ — and by operating-band amplitude ($A > B > D > C$) they are 3.4, 15.0, 8.0 and 8.3 %, likewise not monotone ($p =$

0.97). D therefore lands where the band said it would, but the band spans everything from A's rate to B's, and a single structure at 50 designs has an interval wide enough to contain all three neighbours. The out-of-sample test confirms the two findings that do not depend on r at all — the threshold self-check admits every design, and the pretenders are the geometries the optimizer pushed outside the generator's domain — and leaves the r ordering where Section 3.3 left it: consistent with, not established.

4 Discussion

4.1 Forward scores and downstream use

The transferability diagnostic of [1] — shape correlation r read jointly with the operating-band TMM MAE — answers a forward question: does TMM-pretraining help the surrogate interpolate RCWA? The audit answers a different one: can the optimizer be trusted to commit to the surrogate's argmin? The two come apart because the inverse loop consumes the surrogate's **gradients**, not its average error. A surrogate can have excellent forward MAE — 1.7 to 2.1 % here — and still possess shallow basins, rejected by the reference solver, that the optimizer is built to find; the forward metric never visits those basins, and r , a correlation of *spectra* rather than of error landscapes, cannot see them.

What the data license is narrower than a blanket separation, and worth stating precisely. The **threshold verdict** screened nothing out: all 179 committed designs pass the surrogate's own 5 % check and 16 do not survive the solver, so a pass/fail self-report cannot rank anything, on any structure, at any r . The **continuous** self-report is a different object: on Structure C it correlates with reference-solver error at $\rho = +0.72$ to $+0.83$ across seeds (Sections 3.4 and 3.7), which is exactly the information the thresholded version discards. And the diagnostic itself is not silent on inverse reliability — pretender incidence follows the pre-specified ordering (Section 3.3) — but it orders incidence across three structures at 16 events, which is a direction, not a calibration.

The practical reading is therefore neither "the score licenses downstream use" nor "the score is irrelevant", but: **a forward score, and a thresholded self-report, are not a substitute for evaluating the geometry the loop commits to.** What the audit costs is one solver call per design; what it buys is the difference between 179 designs believed good and 163 that meet the reference solver's tolerance.

4.2 Incidence, severity and the r ordering at this event count

The only quantity that orders with the preliminary r is incidence — oracle success 19 / 18 / 17 along A, C, B at seed 42 and the pooled trend of Section 3.3. Worst-case severity does not: the largest gap is on B (12.1 %), then A (9.9 %), then C (4.7 %), which is monotone under neither convention. Nor does the Δ -flag count (2 / 1 / 7): B carries the most flags because its threshold is the lowest, and the contrast survives a common threshold (2 / 7 / 1 at either A's or B's threshold, B-vs-C Fisher $p = 0.044$ nominal, 0.39 after Holm; `stats_supplement_v9.json`), so it is a real excess of moderate gaps on B rather than an artefact of the scaling alone. A practitioner therefore cannot use r to bound the worst case, and can use it to order incidence only in the weak sense that Section 3.3 establishes at 16 events.

4.3 The pretender phenomenon and offline model-based optimization

That random search manufactures pretenders as readily as gradient descent (Section 3.6) places this work squarely beside the offline MBO / surrogate- gaming literature [13, 14, 15, 16]: optimizing against a fixed learned objective, however obtained, drifts into regions where the objective is over-optimistic. It is the same structure that Goodhart's law [28], reward hacking [29] and reward-model overoptimization [30] describe in other settings. Autofocused oracles [16] and conservative objective models [14] address it by re-training or regularizing the surrogate, and Design-Bench [15] standardises offline-MBO tasks and their evaluation as a benchmark suite. Robust model adaptation [26] instead localises the surrogate around the query point and penalises the local sensitivity of its prediction; our T4 test measures a related finite-scale spectral sensitivity, not RoMA's robustness objective ($K = 100$ Gaussian perturbations at $\sigma = 1$

% of the box width), and finds it small at every pretender — 0/179 designs flagged, mean perturbation MAE 0.79–1.12 % per run, global maximum 2.97 %. The pretenders therefore sit in *smooth* regions of the surrogate at this perturbation scale; that does not show that training with a smoothness regularizer would leave the same optima in place. Our contribution is accordingly orthogonal and diagnostic — a pre-specified accounting of how often the un-regularized, physics-transferred surrogate that practitioners actually ship is wrong, and a demonstration that, within this audit, no tested diagnostic perfectly separated pretenders from confirmed designs on every structure, so the reference solver's own call remains the deciding step.

4.4 Partial signal in the cheap detectors and the solver call as safeguard

The result in Section 3.7 is practically important: the diagnostics a practitioner would reach for first — local smoothness, multi-start agreement, box-edge checks — either fail outright or, like the box-edge flag, carry at most a weak and unusable association (CMH OR 2.18, $p = 0.25$, specificity 0.37), because a pretender is precisely a point where the surrogate is *internally* self-consistent and confident yet *externally* wrong. Internal consistency alone does not certify external accuracy. On Structure C, however, the surrogate's own continuous error and the ensemble quantities do rank the pretenders (Section 3.7), and even there the pass/fail verdict admits every one of them. Two uses of a simulation budget must then be separated (Section 1.3): (i) *surrogate-improving* calls, which retrain or adapt the model where the optimizer is searching [32, 33], and (ii) *certification-only* calls, which leave the model untouched and merely test the committed answer — this paper's protocol, which differs from the iterative model management of [17, 18] in consulting the expensive model once, at the end. This audit quantifies the second. The safeguard this audit supports is the direct one: **put the oracle in the loop at the committed geometry**. One reference-solver call per design converts a silent-failure rate — 8.9 % on this pipeline, 53.1 % on the as-submitted one — into a detected-failure rate: Surr MAE $\leq \tau$ **and** reference-solver MAE $> \tau$, the pretender rule of Section 2.3, not the Δ gap, which is reported separately as severity.

Uncertainty quantification is the obvious missing ingredient, and we tested a piece of it. The standard toolkit — deep ensembles [47] and MC dropout [48] for predictive uncertainty, post-hoc calibration of classifier confidence [49], the benchmark evidence that uncertainty quality often deteriorates under dataset shift and that in-distribution calibration does not ensure it [50], surveyed in [51] — addresses a model asked about inputs unlike its training data. The committed geometries here tend to be such inputs, since the optimizer searches for them and Sections 3.8 and 3.12 find a fifth to a half of them outside the generator's domain. Our detector benchmark (Section 3.7) includes the cheapest usable member of that family, the disagreement of a three-member deep ensemble formed from the three training seeds, together with a held-out-ensemble error and a k -NN distance to the training set. Their AUROCs are reported there: the ensemble quantities reach 0.95–0.98 on C and 0.73–0.80 on B, the training-set distance 0.80 on B but 0.42 on C, and on A none of them is distinguishable from chance. What we have *not* tested is a calibrated Bayesian surrogate trained for the purpose, or a conformal wrapper (for optimizer-induced shift, with the feedback-covariate-shift correction of [59]) — whose coverage guarantee would itself have to be established for designs selected adaptively by the optimizer; that is the natural next experiment and the main qualification on the negative result (W9).

Box 1. Oracle-in-the-loop reliability audit for a learned surrogate in inverse design

1. **Freeze the accounting first:** the threshold τ , the gap factor k , the restart count R , the target count N and the seeds, written down before the main runs. Here $\tau = 5\%$, $k = 2$, $R = 8$, $N = 20$ and the canonical seed is 42. Seeds 123 and 777 were extensions whose recording time relative to their runs is not established (Section 2.7).
2. **Draw the targets from held-out data by one mechanism for every structure.** (2b: *enforce the generator's geometric feasibility constraints where they are known; Section 3.8 shows this reduces but does not remove pretenders, and the unconstrained arm quantifies the cost of omitting them.*)

3. **Run R restarts and commit the lowest final surrogate loss** — the rule a practitioner would actually use.
4. **Call the reference solver once, at the committed geometry** — per design, not per iteration.
5. **Report the τ -anchored pretender count with a Wilson interval as the primary metric** and the gap Δ as severity; do not scale the primary threshold by each surrogate's own forward error.
6. **Treat surrogate-side diagnostics as triage, not as a substitute.**

Cost: one reference-solver call per committed geometry.

4.5 Protocol refinement: τ -anchored accounting over MAE-scaled flags

The $k \times$ (forward test MAE)-scaled Δ -flag measures something other than the pretender count, and the published pipeline shows the two coming apart in the opposite direction from the superseded release. There, Structure A had nine pretenders and no flags because its 9.69 % threshold sat above its own worst Δ — relative scaling under-detected exactly where forward MAE was large. Here the forward MAE has halved, the thresholds have fallen to 3.4–4.2 %, and the flag now **over**-counts: Structure B carries 7 flags against 3 pretenders at seed 42, and A 2 flags against 1, because a Δ of 4 pp trips a 3.44 % threshold while the design still passes the 5 % success check. A threshold tied to the surrogate's own accuracy moves with it, so the same design can be flagged in one release and not the other. The τ -anchored pretender count ($\text{Surr} \leq \tau < \text{RCWA}$) is anchored to the design tolerance instead and is the right primary reliability metric; the Δ -flag is retained as a severity descriptor, not a detector.

4.6 Relation to prior work

The surrogate-gaming *mechanism* itself (committing to the argmin of an imperfect learned objective) is not new — it is well documented in the offline MBO literature [13, 14, 15, 16]. Our novelty is the **pre-specified, multi-structure, multi-seed reliability audit of a physics-transferred surrogate in inverse design, with a reference solver on every valid committed geometry**, and the resulting finding that the forward transferability diagnostic does not supply, by itself, a demonstrated inverse guarantee. The closest recent treatment of the same underlying failure — a neural surrogate losing accuracy outside its training distribution — is the SIMSHIFT benchmark [58], which measures forward-prediction degradation across four industrial simulation tasks under shifts that the benchmark itself specifies, and tests unsupervised domain adaptation as the remedy. This audit asks the downstream version of that question: here the shift is not chosen by the experimenter but produced by the optimizer, which drives the committed geometry to the point the surrogate is most optimistic about (Sections 3.8 and 3.12), and the quantity at stake is the design decision rather than the field error. The two are complementary — the pretenders found here are the natural test cases for the adaptation methods that benchmark evaluates.

Two mappings between that row set and our taxonomy are worth stating. The neural-adjoint *boundary loss* [56] penalises candidates that leave the training-domain box; our **T2** box-edge flag marks a committed geometry within 0.05 of a box face, which is a weaker, interior condition — a design at $u = 0.02$ trips T2 and incurs no boundary penalty — so T2 is a diagnostic neighbour of that regulariser, not its equivalent, and the 9–18 T2-flagged designs per run are an upper bound on what the penalty would have acted on. And re-simulating the top- k neural-adjoint candidates and our Δ **audit** share one operation, a reference-solver evaluation of a surrogate-chosen design, in different roles: there the solver picks the winner among k , here it audits the single winner the surrogate already picked, which is what makes the cost exactly one solver call per design.

4.7 Limitations

- **W1.** Three orderings of the same three structures appear in this literature: the *preliminary* pilot r (A +0.72, C +0.34, B −0.07) that the protocol used and that is the pre-specified convention here;

Table 12: Relation to prior work (setting and failure-mode treatment of the closest studies, with this audit in the final row).

Reference	Setting	Failure-mode treatment
Liu et al. 2018 [52]	NN inverse design, single-fidelity	tandem architecture
Peurifoy et al. 2018 [53]	NN inverse design, single-fidelity	forward + back-prop inverse
Unni et al. 2020 [40]	mixture-density inverse design	multi-modal handling
Jiang & Fan 2020 [54]	simulator/adjoint-in-the-loop generative	always-on simulator validator
Cheng et al. 2024 [55]	TL $\times$ MDN inverse, same fidelity	curriculum within one solver
Ren et al. 2020 [56]	neural-adjoint inverse benchmark	boundary loss keeps the search inside the training domain; top- k candidates re-simulated
Jiang & Fan 2019 [33]	simulator-in-the-loop generative design	solver inside the training objective
Ovadia et al. 2019 [50]; Abdar et al. 2021 [51]	UQ under dataset shift	calibrated predictive uncertainty as the guard
Setinek et al. 2025 [58]	neural-surrogate benchmark under distribution shift, forward prediction (four industrial simulation tasks)	shift fixed by the benchmark's train/test split; unsupervised domain adaptation as the remedy
Kumar & Levine 2020 [13]	offline model-based optimization	direct model inversion (objective value to design)
Trabucco et al. 2021 [14]	offline model-based optimization	conservative objective models
Fannjiang & Listgarten 2020 [16]	offline model-based optimization	autofocused oracles
The forward study [1]	cross-fidelity TL, forward	joint r /operating-band-MAE diagnostic; controlled negative-transfer tests; a preliminary inverse-failure probe (its Supplementary S13)
This audit	cross-fidelity TL, inverse , 3 structures $\times$ 3 seeds	pre-specified Δ audit + measured 4-type taxonomy + uniform-random nulls + r non-certification + mechanism control

the printed r of [1] (B +0.96, A +0.83, C +0.65); and the printed operating-band TMM MAE (A 7.9 %, B 8.9 %, C 16.9 %), which is the component the companion says the forward benefit actually follows. The published values come from the regenerated data-generation pipeline audited here, whereas the preliminary values came from the pilot pools of the as-submitted release, so the conventions are not three readings of one dataset. All three are reported (Section 3.3); none is confirmed at the pre-specified seed, and only the preliminary ordering reaches nominal significance in the exploratory pooled trend.

- **W2.** All three structures sit on the same 400–1800 nm grid in the published pipeline, so the band mismatch that constrained cross-structure comparison in the as-submitted release (A on 380–780 nm against B and C on 400–1800 nm) does not apply here; Section 3.10's legacy arm still carries it, and the comparisons there stay dimensionless for that reason. The design boxes still reach $P/\lambda_{\min} = 1.5$ (A) and 2.0 (B, C), so several diffraction orders are open at the short-wavelength end on every structure.
- **W3.** $N = 20$ /structure/seed (D: 50, one seed); counts have wide CIs, so we use "not supported" rather than "refuted," and report effective- N where T3 clustering is present. The three seeds of a structure share the same 20 targets, so pooled counts (179) are correlated and their nominal Wilson intervals are optimistic (Section 3.4).
- **W4.** Single architecture, single physics family (MIM, Cr); scope is the release audited here, and the paper is positioned as a companion to [1].
- **W5.** The optimizer compresses self-reported MAE into 0.06–4.6 %, which attenuates rank correlations; this range restriction is part of the phenomenon, not a nuisance to be corrected away.
- **W6.** The two Structure-C quirks of the as-submitted release — a fine-tune pool of $n_{\text{train}} = 300$ labelled " $n = 350$ ", and a checkpoint normalization box of $W_x, W_y \in [50, 400]$ nm against a dataset sampling those parameters to ~ 720 nm, which put a third of the committed C geometries outside $[0, 1]$ — are **absent from the published pipeline audited here**: C trains on 350 of 487 reliable samples with the corrected bounds and the 18-feature set. They remain properties of the legacy arm of Section 3.10 and are one of the several differences that arm bundles together.
- **W7.** The from-scratch control (Section 3.6) bounds, rather than removes, this concern. Over the 60 designs it shares with the transfer-learned run at the pre-specified seed the paired shift in Δ is +0.14 pp $[-0.07, +0.44]$ and the pretender counts are 6 against 4; over the 179 designs of all three seeds the target-clustered shift is +0.20 pp $[+0.03, +0.38]$ and the counts are 16 against 17, so a surrogate trained without transferred weights produces pretenders at a rate this design cannot distinguish from the transfer-learned one. The control is not identical in accuracy — on A its forward MAE is 2.57–2.62 % against 1.68–1.78 % at every seed and its Δ shifts by +0.82 pp $[+0.55, +1.32]$ pooled, on B and C the pooled shifts are +0.02 and +0.01 pp — so it bounds the pooled contribution of transfer to a few tenths of a percentage point of Δ , and A's own to about +0.5 to +1.3 pp, rather than showing either to be zero. What it does not establish is the behaviour of a much *less* accurate surrogate, or of one carrying a different kind of certificate.
- **W8.** The reference solver is the same coupled-wave code that generated the training labels, so surrogate and oracle share a systematic error, and this is a reference-fidelity statement rather than a convergence-certified one. The published pipeline chooses the Fourier order per wavelength from $\{9, 13, 17\}$ instead of fixing it at 5, which reduces the discrepancy relative to that setting that the companion's convergence study ([1], Supplementary S16) measured at up to ≈ 6 pp on A and ≈ 15 pp on C at long wavelengths; the order-7 revalidation that mitigated it in the legacy arm is therefore superseded rather than repeated here. The 64×64 real-space raster used to sample every geometry is a second fidelity setting: its Fourier couplings are free of index wrap-around only up to $N = 15$, so at $N = 17$ the outermost couplings of the truncated operator (≈ 2 % of its

entries) fold. Re-solving 54 committed designs on a 256×256 raster at the pinned order — every design within 2 pp of τ (42), the three with the largest predicted raster sensitivity, and ten drawn at random, across all four structures and every training seed; 53 at all 100 wavelengths and 1 on 10–20-wavelength subsamples — changed no pretender verdict (mean $|\Delta A|$ 0.01–1.20 pp per design, worst single wavelength 2.2 pp, against $\tau = 5$ %); this bounds the sampled designs, which include every design within 2 pp of τ , not the population. The companion justified this raster for feature sizes ≥ 50 nm; Structure B's design space extends to 10 nm, and 471 of the 500 samples behind its printed tables (94.2 %) contain a feature below 50 nm, 309 (61.8 %) below two pixels (`paper1_b_feature_census_v11.json`), so for that structure the premise does not hold. Re-solving 30 of the companion's Structure-B dataset samples, stratified by feature size, on the same 256 raster moves their labels by 0.06–0.90 pp (mean $|\Delta A|$ per sample; worst single wavelength 2.6 pp), with the grid-64 control reproducing the archived labels to 0.002 pp (`grid_delta_summary_v11.json`). The oracle also applies no admissibility filter: 2 of 289 archived committed-design spectra carry one point below the companion's -0.005 floor (the grazing-order artefact of [1], Supplementary S16), and neither verdict changes if the point is clipped or dropped (`oracle_admissibility_v11.json`). The solver still describes unconverged cases at its order cap, so nothing here certifies physical truth. Section 3.11 re-solves the committed geometries, on the same raster, at a fixed order 5 and finds that the order alone doubles the pretender count on identical, published-pipeline designs ($6 \rightarrow 13$ of 60): the verdict on *these* designs is sensitive to the audit's own truncation, though not, on this evidence, to its raster. The probe was not run on the as-submitted arm's own (different) committed geometries, so how much of its 53.1 % owes to truncation versus to its other differences (Section 3.10) is not established.

- **W9.** The surrogate-side diagnostics benchmarked here are the cheap ones a practitioner already has: a three-seed ensemble, held-out ensemble error, k -NN distance to the training set, TMM disagreement and the four taxonomy tests. We did **not** train a calibrated Bayesian or conformal surrogate for the purpose, so the negative result is scoped to the diagnostics tested rather than to uncertainty quantification as a whole.

5 Conclusion

We pre-specified a reliability accounting protocol and used it to ask whether a forward transferability diagnostic certifies the inverse usability of a physics-transferred neural surrogate. Run on the published pipeline of [1] — three MIM structures spanning the diagnostic's range, three training seeds each, the reference solver called at every valid committed geometry — the answer is that the surrogate's own **pass/fail check screened out none of the designs the solver rejects**: all 179 valid committed designs pass it and the solver rejects **16 (8.9 %; [4.0, 14.5] target-clustered, [5.6, 14.0] nominal)** — of 180 committed designs, 163 confirmed, 16 rejected and 1 unresolved by the solver — a count that rises to 31 % of those still certified if the tolerance is tightened to 2.5 %, and a fourth structure adds 4 of 50 to the same pattern.

Three controls show what is not necessary. A surrogate trained from scratch produces pretenders at a rate this study cannot distinguish from the transfer-learned one, confining the optimizer to the generator-feasible region reduces the count without removing it, and budget-matched random search produces them as well, at counts the study cannot rank against gradient descent: none of the three isolates a single cause, and what every one of them keeps is committing to the argmin of a surrogate that is imperfect precisely where it is most optimistic. Running the identical protocol on the release this pipeline superseded gives 95 of 179 (53.1 %) — the same structures, thresholds and target-selection rule, a sixfold difference — so a reliability claim of this kind describes a release rather than a method, and the release worth auditing is the one being shipped.

Two findings cut the other way and are reported with their limits rather than buried: pretender incidence follows the pre-specified r ordering across these structures, and on one structure the surrogate's continuous error estimate separates pretenders from confirmed designs almost perfectly. Both rest on event counts too

small to build a rule on, which is itself the useful statement: at these rates, a study of this size can bound the effect but not calibrate a threshold. A fourth structure audited out of sample lands inside the rate band recorded for it (4 of 50) without making the ordering monotone, and repeats the regularity of B and A: its pretenders are designs committed outside the generator's feasible region. C remains the counterexample — all five of its pretenders are feasible — so that regularity is a property of three of the four structures, not a law.

The deliverable is the protocol and what it costs. One reference-solver call per committed geometry — 103 to 4595 s here, against a structure-median 1.5–2.3 min of optimization per design (wall clock from the run logs) — converts a set of designs believed good into a set that meets the reference solver's tolerance, and Box 1 states the procedure as six steps.

Code, Data, and Compute

A single structure-agnostic codebase (PyTorch [57]) runs every stage for all four structures: fine-tuning (strict load of TMM-pretrained checkpoints regenerated with the recipe of [1]), inverse design (normalized-variable Adam with eight restarts), reference-solver validation (torcwa 0.1.4.2 [46]), reliability accounting, the cross-structure synthesis, the two mechanism controls (budget-matched random search and single-start descent), the from-scratch and feasibility-constrained controls, the fixed-order cross-solver probe and the Structure-D audit. Per-run artifacts for every structure $\times$ seed, the aggregated evidence files behind every number in Section 3, and the figure script that regenerates all figures from those files without recomputation are in the repository (Data availability). Compute: one NVIDIA RTX 4070 Ti SUPER; reference-solver time per committed geometry 103–4595 s at the printed pipeline's per-wavelength adaptive order (median 1655 s; B 158 s, C 1662 s and A 3485 s at the median), against 16–39 s for the as-submitted release's fixed order 5. All simulation code is vendored at upstream commit `cb486b5` and the exact input hashes are listed in the repository's manifests (Data availability). The raw data of [1] are at github.com/BigMountain87/PBTL-MIM; that release, at the audited commit, shipped no checkpoints, so the regenerated ones are in this paper's archive.

Acknowledgements

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Generative AI tools were used in preparing this work: Claude (Anthropic; Opus 4.8, Opus 5 and Fable 5.1) assisted with drafting, code refactoring and analysis scripts, and Codex (OpenAI; gpt-6-astra) assisted with checking code, artifacts and manuscript claims. The authors take responsibility for the experiments, the protocol, the claims and their interpretation; the review transcripts are in the repository.

Data availability

The code and artifacts behind this study are openly available at github.com/BigMountain87/PBTL-MIM-audit (tag `v1-preprint`); a Zenodo archival deposit of the same release is in preparation. The repository contains the code, the pre-specified protocol, the per-run artifacts for every structure and training seed, the aggregated evidence files behind every number reported here, and the figure scripts. It also contains the TMM-pretrained checkpoints the surrogates are fine-tuned from (regenerated for A, B, C and D with the recipe of [1], whose public release at the audited commit shipped no checkpoints), the as-submitted arm's inputs and artifacts, and the Structure-D dataset. The raw datasets of [1] are additionally available at github.com/BigMountain87/PBTL-MIM.

Conflict of interest

The authors declare no competing interests.

Author contributions

S.-B. Choi: conceptualization, methodology, software, formal analysis, investigation, data curation, writing (original draft), visualization. J.-M. Choi: validation, writing (review and editing). J. Kim: methodology, validation, writing (review and editing). C.-M. Kang: supervision, resources, project administration, writing (review and editing).

ORCID iDs

To be added at publication.

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